

Maximum Tsallis Entropy Distributions for Robust and Efficient Sparse Learning from Correlated Data

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Abstract

This paper addresses the limitations of Gaussian distribution assumptions in statistical sparse learning, particularly in modeling correlated and heterogeneous data. Conventional Gaussian models often lack robustness towards outliers and underlying distribution assumptions. To overcome these limitations, we propose the use of the q Gaussian distribution, derived from Tsallis entropy maximization, as a robust alternative. This is notably relevant in biostatistics, where the presence of correlated observations and heterogeneity, such as in genetic and longitudinal studies, is prevalent. Our contributions include modeling of correlated data through the re-derived multivariate probability density function from Tsallis entropy maximization, thereby addressing the limitations inherent in conventional Gaussian models. Furthermore, we introduce a novel framework that adapts numerical methods designed to find equilibria in flows to tackle composite optimization problems prevalent in statistical sparse learning. Applying this framework to the Hager-Zhang conjugate gradient algorithm [23], we develop a numerically stable and efficient algorithm for sparse statistical learning. The q Gaussian distribution, informed by the principle of maximizing Tsallis entropy, presents a viable and flexible alternative to Gaussian-based methods. This paper not only contributes to the theoretical understanding of statistical distributions and optimization techniques, but also paves the way for practical data analysis.

Keywords: Statistical Learning, high-dimensional, Entropy, Robust, Optimization

1 Introduction

In the realm of statistical sparse learning, the pursuit of robust and efficient methodologies remains paramount, especially when confronted with the complexities of correlated data. The principle of maximizing Shannon's entropy stands as a pivotal framework that has led to the derivation of nearly all frequently utilized statistical distributions to date [12]. This principle's application has notably revealed that the multivariate Gaussian distribution

maximizes Shannon’s entropy under the first two moments, significantly influencing the landscape of statistical sparse learning. The Gaussian assumption has become a fundamental cornerstone of numerous statistical sparse learning problem formulations, and its core assumptions are rarely re-examined or challenged.

However, the Gaussian distribution’s features, particularly its exponential tail decay and the absence of a shape parameter, can present substantial limitations. Specifically, the lack of robustness towards outliers and a limited capacity to accurately represent the distribution’s shape results in violations of the Gaussian assumption in statistical modeling. Such violations have many practical repercussions, including the potential for erroneous Type I error rates and the lack of robustness towards distribution shape when estimating the dispersion parameter, motivating the development of alternative approaches.

Dispersion or volatility parameters encapsulate critical and often decisive information about distributions. Their estimations, specifically in transformations of predicted outcomes, are often indispensable. For instance, in the context of log-normal distributions, the mean is directly influenced by the volatility parameter derived from the underlying Gaussian distribution. Likewise, principles like the Law of the Unconscious Statistician (LOTUS), which rely on accurate estimation of volatility and precise understanding of the distribution’s shape, highlight the importance of determining this parameter for dependable prediction and statistical modeling.

Volatility estimation is of great importance in finance. Specifically, Itô’s lemma, often used in stochastic calculus for option pricing, explicitly highlights the importance of volatility’s contribution. Delving into the realm of stochastic calculus, Itô’s lemma provides a mathematical framework that elegantly captures volatility’s impact on dynamic systems. Itô’s lemma states that for a twice-differentiable function f ,

$$df(t, X_t) = \left(\frac{\partial f}{\partial t} + \mu \frac{\partial f}{\partial x} + \frac{1}{2} \sigma^2 \frac{\partial^2 f}{\partial x^2} \right) dt + \sigma \frac{\partial f}{\partial x} dW_t, \quad (1)$$

where the term $\sigma^2 \frac{\partial^2 f}{\partial x^2}$ specifically denotes the contribution of volatility to changes in the function f . This mathematical representation is pivotal in finance, where the phenomenon, termed *volatility smile*, challenges the foundational assumptions of the Black–Scholes–Merton (BSM) model, signaling empirical deviations from expected normality in option pricing models. These deviations have propelled the exploration of alternative distributions capable of more accurately reflecting market realities [42].

In response to these limitations of Gaussian distributions, the q Gaussian distribution, derived from the maximization of the Tsallis entropy, emerges as a compelling alternative. The q Gaussian distribution is celebrated for its flexibility in modeling the diverse shapes of bell-curved distributions, including the ability to account for heavy-tailed distributions — a feature crucial for robust modeling of financial returns. It provides a more accurate representation of financial returns on platforms such as the *New York Stock Exchange (NYSE)* and *National Association of Securities Dealers Automated Quotations (NSDAQ)*

[5, 6, 15]. Despite its proven advantages in finance, the incorporation of Tsallis entropy-maximizing distributions within the domain of statistical sparse learning and biostatistics remains limited. To the best of our knowledge, this paper represents the initial endeavor to apply Tsallis entropy-maximizing distributions for biostatistical data modeling. Correlated observations, frequently encountered in genetic and longitudinal studies [18, 48, 14], as well as heterogeneity of the variance, will be specifically addressed by our proposed Tsallis entropy-maximizing model for correlated data.

Hence, this paper advocates for the application of the q Gaussian distribution in modeling correlated data within sparse statistical learning frameworks. Our approach relaxes the conventional reliance on normality assumptions; We aim to demonstrate that the intricate characteristics of q Gaussian distributions can profoundly enhance the modeling of correlated data, offering a robust and versatile alternative to conventional Gaussian-based methods.

Maximum likelihood estimation is one of the most commonly used estimation techniques. However, the estimation process encounters notable computational obstacles when dealing with high-dimensional and extra-large datasets. Oracle penalties, favored for their efficacy in facilitating variable selection, present an attractive yet complex solution to sparse learning problems. However, oracle penalties are notable for their nonconvex and nonsmooth nature [40], which lead to considerable optimization challenges. Recently, proximal methods have demonstrated an unmatched speed of convergence, thereby surpassing most other approaches in efficiently handling estimation in nonsmooth problems [26]. Simultaneously, the Krylov subspace method, recognized among the top ten algorithms for computing in science and engineering of the twentieth century, lays a solid foundation for numerical analysis. The conjugate gradient method, a prominent member of the Krylov subspace methods, has been applied extensively in various areas and is a fundamental numerical tool in solving partial differential equations [41]. While the nonlinear conjugate gradient performs exceptionally well in terms of its convergence speed and numerical stability, much better than accelerated gradient or gradient descent, its global convergence depends on the line-search step, whereas accelerated gradient and gradient descent methods do not necessarily require the line search step to achieve global convergence [19, 63]. Motivated by these methodologies, our paper introduces a proximal conjugate gradient method that can be applied to solve q Gaussian sparse learning problems. This method aims to effectively combine the theoretical strengths of both proximal methods and Krylov subspace techniques. Additionally, our paper addresses the line search step needed for the proximal nonlinear conjugate gradient method to establish global convergence.

We re-derive the probability density function for the multivariate q Gaussian distribution from a Tsallis entropy maximizing perspective in Lemma 1. This derivation allows for a nuanced understanding and application of this model in statistical analysis. Furthermore, *our contributions in this paper are as following:*

1. *We apply the derived density to model correlated and heterogeneous data effectively, while carrying out the sparse statistical learning at the same time.*
2. *Sparse statistical learning involves minimizing a composite optimization problem, aimed at minimizing a composite objective function composed of a globally Lipschitz-smooth term, which may be nonconvex, and a convex nonsmooth term. A variety of numerical methods are available to find equilibrium points for globally Lipschitz flows. By employing the Moreau envelope and linearizing the smooth term, we develop a framework that allows any numerical method designed for finding equilibrium points in globally Lipschitz flows to be adapted into a numerical optimization algorithm for minimizing the composite objective function.*
3. *Leveraging the framework introduced above, we implement it with the state-of-the-art Hager-Zhang conjugate gradient method [23]. This implementation yields a proximal conjugate gradient algorithm that is not only computationally efficient but also numerically stable, suitable for a wide range of statistical sparse learning challenges. This includes the robust sparse learning approach we devised based on the concept of maximizing the Tsallis entropy distribution.*

The structure of the paper is organized as follows:

Section 2 delves into the foundational properties of Tsallis entropy, drawing upon previous literature to establish a comprehensive background. Following this, Section 3 introduces the concept of q -moments. This section then elaborates on employing Tsallis entropy maximizing distribution to effectively model the q -correlation structure. In Section 3, we also re-derive the probability density function maximizing Tsallis entropy under the first and second central q -moment constraints, incorporating all relevant parameters for a likelihood-based approach to statistical analysis.

Our discussion transitions to the challenges and strategies of optimization in Section 4. This section is twofold; initially, in Section 4.1, we present essential background knowledge from variational and nonsmooth analysis. This foundation is critical for our novel contribution: the development of a proximal framework to transform any first-order numerical optimization algorithm to a proximal counterpart by leveraging the properties of the Moreau envelope, detailed in Section 4.2. In Section 4.3, we apply this innovative framework to the state-of-the-art Hager-Zhang conjugate gradient algorithm. This adaptation produces a proximal version for tackling sparse statistical learning challenges. The efficacy of this method is further showcased in Section 5, where we outline the application of our proximal Hager-Zhang conjugate gradient algorithm to optimize a penalized q Gaussian likelihood function. This section also lays out a map from problem formulation to the practical aspects of prediction using models trained with our approach. Finally, Section 6 synthesizes our contributions, offering a reflective conclusion and proposing avenues for future research.

2 Tsallis Entropy

For an arbitrary random variable X , Shannon's Entropy [50] poses the definition

$$H(X) := -\mathbb{E} \log(p(X)) = - \int \log(p(x)) d\mu_X, \quad (2)$$

where p is the likelihood function for X . Over a given (likelihood) function space

$$\mathcal{P} := \left\{ p(x) \mid \forall x \in \mathcal{X}, p(x) \geq 0, |\mathbb{E} \log(p(X))| < \infty \text{ and } \int_{\mathcal{X}} 1 d\mu_X = 1 \right\},$$

the Shannon's entropy is a *strictly concave* function, which implies uniqueness of the maximizer. Many commonly-used distributions have been shown to maximize Shannon's entropy under certain given constraints [12]. For example, uniform distribution, whether in discrete or continuous case, are to maximize (2) over a compact support, with open sets defined by discrete or Euclidean topology, respectively. The exponential distribution is defined as maximizing (2) over $\mathbb{R}_{\geq 0}$ and with a constraint that the first moment is a constant, $\frac{1}{\lambda}$; where λ later turns out to be the scale parameter. And the Gaussian distribution maximizes (2) over $\mathbb{R}$ with given mean and variance. More examples can be given. For example, the constraint to obtain a Laplace distribution is a given mean absolute deviation, etc.

Additivity is a key element of Shannon's entropy. That is, let A_1, A_2 be two independent event sets, then the information of the intersection $I(\mathbb{P}(A_1 \cap A_2)) = I(\mathbb{P}(A_1) \cdot \mathbb{P}(A_2)) = I(\mathbb{P}(A_1)) + I(\mathbb{P}(A_2))$ — such homomorphism was considered particularly useful in Shannon's view [50]. Later in the 1980s, Tsallis constructed an entropy similar to Shannon's entropy but without the additivity property. To see how Tsallis' entropy was developed, first we look at Tsallis' q -exponential function, which is defined as $\exp_q : \mathbb{R} \mapsto \mathbb{R}$, given by

$$\exp_q x := \begin{cases} ((1 + (1 - q)x))^{\frac{1}{1-q}}; & 1 + (1 - q)x > 0 \\ 0; & \text{else} \end{cases} \quad (3)$$

For $q > 1$, $\exp_q$ is bijective over $(0, \frac{1}{q-1})$. The inverse function, called the q -logarithmic function, is given by

$$\ln_q x := \frac{x^{1-q} - 1}{1 - q}. \quad (4)$$

Based on this deformed q -exponential function, [56] developed *Tsallis entropy* by replacing the log function in 2 with q -log function (4) and replacing the expectation with q -expectation operator [56]:

$$S_q(X) = - \int_{\mathcal{X}} p^q(x) \ln_q p(x) dx =: -\mathbb{E}_q \ln_q p(X) \quad (5)$$

$$= \frac{1}{q-1} \left(1 - \int_{\mathcal{X}} p^q(x) dx \right) \quad (6)$$

where $q \in \mathbb{R} \setminus \{1\}$ is a constant, and $\mathbb{E}_q f(X) := \int_{\mathcal{X}} f(x) \cdot p^q(x) dx = \langle f(x), (d\mu_X)^q \rangle$ is referred to as the q -expectation operator. Tsallis entropy is also known as *non-extensive* entropy; namely for arbitrary independent two random variable X_1, X_2 :

$$S_q(X_1, X_2) = S_q(X_1) \oplus_q S_q(X_2), \quad (7)$$

where “ $\oplus_q$ ” is defined as $\forall a, b \in \mathbb{R}$,

$$a \oplus_q b := a + b + (1 - q) ab. \quad (8)$$

Expectation has been used to characterize statistical distributions. However, one significant drawback of the expectation (linear) operator is the lack of continuity for some distributions; such as the Cauchy distribution. Therefore, the q -expectation operator, $\mathbb{E}_q$, provides robustness when characterizing the distributions in the real domain. If the tail of the function vanishes at a rate of $O((\log x)^{-1})$, the function will not have a proper integral if the support is unbounded. Thus, for any distribution whose likelihood function is bounded in uniform norm, $\exists q \in \mathbb{R}_{>0}$ such that $\mathbb{E}_q$ is continuous at the likelihood function in the function space we are considering.

3 Tsallis Entropy Maximizing Distribution to Accommodate the q -Correlation Structure

The Gaussian distribution maximizes the Shannon’s entropy in the following problem:

$$\begin{aligned} \max_{\phi \in \mathcal{P}} & - \int_{\mathbb{R}^n} \phi(x) \log(\phi(x)) dx \\ \text{s.t. } & \phi \geq 0; \\ & \int_{\mathbb{R}^n} \phi(x) dx = 1; \\ & \int_{\mathbb{R}^n} x \cdot \phi(x) dx = 0; \\ & \int_{\mathbb{R}^n} xx^T \cdot \phi(x) dx = \Sigma; \end{aligned} \quad (9)$$

for some $n \in \mathbb{N}_+$ and $\Sigma \in \mathbb{R}^{n \times n}$, $\Sigma \succ 0$. For the sake of parsimony, in (9) we assume that the distribution is centered. To set the central trend parameter, or the mean parameter in the specific case of the Gaussian distribution, the likelihood function ϕ can be simply translated $x \mapsto x - \mu$ to incorporate the parameter μ for the central trend. Entropy functions are invariant under translation.

Similarly to how the multivariate Gaussian distribution maximizes Shannon's entropy in a Euclidean space, the multivariate q Gaussian distribution maximizes Tsallis entropy in a Euclidean space. Specifically, the optimization problem is formulated as:

$$\max_{\phi \in L^q(\mathbb{R}^n)} - \int_{\mathbb{R}^n} \phi^q(x) dx \quad (10)$$

$$\text{s.t. } \phi \geq 0;$$

$$\int_{\mathbb{R}^n} \phi(x) dx = 1; \quad (11)$$

$$\frac{\int_{\mathbb{R}^n} x \cdot \phi^q(x) dx}{\int_{\mathbb{R}^n} \phi^q(x) dx} = 0; \quad (12)$$

$$\frac{\int_{\mathbb{R}^n} xx^T \cdot \phi^q(x) dx}{\int_{\mathbb{R}^n} \phi^q(x) dx} = \Sigma, \quad (13)$$

where $q > 1$. The feasible set of Lebesgue space $L^q(\mathbb{R}^n)$ is to ensure the well-definedness of Tsallis entropy. The normalization constraint (11) implies that $\phi \in L^1$; however, $\phi \in L^1$ does not imply $\phi \in L^q$, as the embedding property $L^1 \subseteq L^q$ fails to hold for Lebesgue measure on $\mathbb{R}^n$. As an example, consider the one-dimensional example of the probability density function

$$\tilde{\phi}(x) = \begin{cases} \frac{1}{4} |x|^{-\frac{1}{2}} & \text{for } x \in (-1, 1) \setminus \{0\}; \\ 0 & \text{else.} \end{cases} \quad (14)$$

Clearly, $\tilde{\phi} \in L^1$ but $\tilde{\phi} \notin L^2$. Note that (12) and (13) are the first and second moment constraints using the q -expectation operator $\mathbb{E}_q$. As noted by [34], maximizing any member of the generalized class of power-law entropies, including Renyi entropy, Havrda and Charvat entropy, Arimoto entropy, and Tsallis entropy, all yield the identical power-law objective function (10). Regarding the constraints, $\int_{\mathbb{R}^n} \phi^q(x) dx$ is the normalization factor for the q -expectation. [34] further noted that optimizing the problem formulated above is equivalent to the following problem:

$$\max_{\varphi \in L^s(\mathbb{R}^n)} \int_{\mathbb{R}^n} \varphi^s(x) dx \quad (15)$$

$$\text{s.t. } \varphi \geq 0;$$

$$\int_{\mathbb{R}^n} \varphi(x) dx = 1;$$

$$\int_{\mathbb{R}^n} x \cdot \varphi(x) dx = 0;$$

$$\int_{\mathbb{R}^n} xx^T \cdot \varphi(x) dx = \Sigma.$$

In (15), $s := q^{-1} \in (0, 1)$, thus $L^s(\mathbb{R}^n)$ is a quasi-normed space; $\varphi(x) := \frac{\phi^q(x)}{\int_{\mathbb{R}^n} \phi^q(x) dx}$. If the maximizer of (15) is φ , then the maximizer of (10), ϕ , will be normalized

$$\phi(x) \propto \varphi^{1/q}. \quad (16)$$

Several important properties were proposed previously regarding the q Gaussian distributions in previous studies [57, 11]. Notably,

1. Using Bregman information divergence, Problem (15) has a unique maximizer of the form

$$\varphi(x; s) = A_s \left(1 - (s - 1) \beta' \langle x, \Sigma^{-1} x \rangle\right)_+^{\frac{1}{s-1}} \quad (17)$$

for some $s \in \left(\frac{n}{n+2}, \infty\right) \setminus \{1\}$, normalization constant A_s , and some dispersion parameter β' .

2. If $X \sim q\text{Gaussian}(q, \Sigma)$, $H \in \mathbb{R}^{\tilde{n} \times n}$ and $\text{rank}(H) = \tilde{n}$. Then $\tilde{X} \sim q\text{Gaussian}(\tilde{q}, H\Sigma H^T)$ with

$$\frac{2}{1 - \tilde{q}^{-1}} - \tilde{n} = \frac{2}{1 - q^{-1}} - n. \quad (18)$$

3. If X_1, X_2 are both q Gaussian random vectors but independent, a linear combination of $H_1 X_1 + H_2 X_2$ is *not* q Gaussian.
4. The duality property: if $X \sim q\text{Gaussian}(q, \Sigma)$ with $1 < q < 1 + \frac{2}{n}$, let the degree of freedom for X be $m := \frac{2}{q-1} - n$ and $\Lambda := m\Sigma$, then

$$\begin{aligned} \frac{X}{\sqrt{1 - \langle X, \Lambda^{-1} X \rangle}} &\sim q\text{Gaussian}\left(\tilde{q}, \frac{m}{m+4}\Sigma\right) \\ \text{with } \frac{1}{\tilde{q}^{-1} - 1} &= \frac{1}{1 - q^{-1}} - \frac{n}{2} - 1, \end{aligned}$$

and $0 < \tilde{q} < 1$.

Property 1 will be used in our Lemma 1. Property 2 implies that any components of a q Gaussian random vector are also q Gaussian, while Property 3 implies that two independent q Gaussian vectors are not jointly q Gaussian.

By the equivalence of problems (10) and (15) discussed before, (17) can be rewritten as

$$\phi(x; q, \Sigma) = (\alpha - \beta \langle x, \Sigma^{-1} x \rangle)_+^{\frac{1}{1-q}} \quad (19)$$

for some constant (parameter) $\alpha, \beta, q \in (0, 1 + \frac{2}{n}) \setminus \{1\}$; $x_+ := \max(0, x)$. As shown later in the proof of Lemma 1, the dimension-related upper bound $1 + \frac{2}{n}$ is due to the normalization constraint (11). When $0 < q < 1$, the density represents a distribution with bounded support; when $q > 1$, the density is a generalization of the bell curve distributions,

and with $q \searrow 1$ the Gaussian distribution is recovered. A higher value of q corresponds to heavier tails in shape. The duality between the q Gaussian random vectors with $0 < q < 1$ and $1 < q < 1 + \frac{2}{n}$ was given by [57], which we discussed in Property 4 in Section 3. Distributions with bounded support correspond to $0 < q < 1$, and distributions with heavy tails correspond to $1 < q < 1 + \frac{2}{n}$. For the scope of this paper, we will focus only on the heavy-tail distributions; i.e., the case when $q > 1$. [60] derived the q Gaussian probability density function for $1 < q < \frac{n+4}{n+2}$, when the multivariate q Gaussian density becomes the scaled density of the multivariate student's t distribution. To incorporate the case of $q \in [1 + \frac{2}{n+2}, 1 + \frac{2}{n})$, when the variance does not exist but the q -variance can be used to capture the volatility/dispersion of the data, [58] also derived the resulting density; however, since a typo was found in that paper, we re-derive the density in Lemma 1. The parameters presented in the density formula (20) are of particular interest to statisticians, as parameter inference is the key to statistical analysis and prediction. The case of $q \in [1 + \frac{2}{n+2}, 1 + \frac{2}{n})$ will allow the resulting q Gaussian distribution to incorporate the wider class of distributions without finite moments but finite q -moments; such as the Cauchy distribution. Therefore, modeling using the q Gaussian distribution with q allowed to take the value in $[1 + \frac{2}{n+2}, 1 + \frac{2}{n})$ will be more robust.

Lemma 1. *When $q \in (1, 1 + \frac{2}{n})$, the unique solution to (10) is:*

$$p(x; q, \Sigma) = \frac{1}{|\pi\Sigma|^{1/2}} \cdot \frac{\Gamma\left(\frac{1}{q-1}\right)}{\Gamma\left(\frac{1}{q-1} - \frac{n}{2}\right)} \cdot \left(\frac{2}{q-1} - n\right)^{-\frac{n}{2}} \cdot \left(1 + \left(\frac{2}{q-1} - n\right)^{-1} \cdot \langle x, \Sigma^{-1}x \rangle\right)^{\frac{1}{1-q}}. \quad (20)$$

Proof. By (17) and the equivalence of the problems (10) and (15), let the solution to (10) be denoted by

$$p(x; q, \Sigma) = \frac{1}{Z} (\gamma + \langle x, \Sigma^{-1}x \rangle)^{\frac{1}{1-q}} \quad (21)$$

for some $Z, \gamma > 0$. Feasibility for problem 10 when $q \in (1, 1 + \frac{2}{n})$ was given in [57]. Hence, the strictly concavity of the objective function 10 implies that the optimal solution is unique. The symmetry of $p(x; q, \Sigma)$ is implied by (21); thus, we reformulate the problem (10) as the following equivalent problem:

$$\begin{aligned} \max_{p \in L^q(\mathbb{R}^n)} & - \int_{\mathbb{R}^n} (p(x; q, \Sigma))^q dx \\ \text{s.t. } & p(x; q, \Sigma) \geq 0; \\ & \int_{\mathbb{R}_{>0}^n} p(x; q, \Sigma) dx = 2^{-n}; \\ & \frac{\int_{\mathbb{R}^n} x \cdot p^q(x; q, \Sigma) dx}{\int_{\mathbb{R}^n} p^q(x; q, \Sigma) dx} = 0; \end{aligned}$$

$$\frac{\int_{\mathbb{R}^n} x x^T \cdot p^q(x; q, \Sigma) dx}{\int_{\mathbb{R}^n} p^q(x; q, \Sigma) dx} = \Sigma. \quad (22)$$

Thus,

$$\begin{aligned} Z &= 2^n \int_{\mathbb{R}_{>0}^n} (\gamma + \langle x, \Sigma^{-1} x \rangle)^{\frac{1}{1-q}} dx \\ &= 2^n |\Sigma|^{1/2} \int_{\mathbb{R}_{>0}^n} (\gamma + \langle x, x \rangle)^{\frac{1}{1-q}} dx \\ &= 2^n |\Sigma|^{1/2} \int_0^\infty r^{n-1} (\gamma + r^2)^{\frac{1}{1-q}} \left(\prod_{i=1}^{n-2} \int_0^{\frac{\pi}{2}} \sin^{n-1-i}(\theta_i) d\theta_i \cdot \int_0^{\frac{\pi}{2}} 1 d\theta \right) dr \\ &= 2^n |\Sigma|^{1/2} \cdot \left(\prod_{i=1}^{n-2} \int_0^{\frac{\pi}{2}} \sin^{n-1-i}(\theta_i) d\theta_i \cdot \int_0^{\frac{\pi}{2}} 1 d\theta \right) \cdot \int_0^\infty r^{n-1} (\gamma + r^2)^{\frac{1}{1-q}} dr \\ &= \pi 2^{n-1} |\Sigma|^{1/2} \cdot \left(\prod_{i=1}^{n-2} \frac{1}{2} \frac{\Gamma\left(\frac{n-i}{2}\right) \sqrt{\pi}}{\Gamma\left(\frac{n-i+1}{2}\right)} \right) \cdot \int_0^\infty r^{n-1} (\gamma + r^2)^{\frac{1}{1-q}} dr \\ &= 2\pi^{\frac{n}{2}} |\Sigma|^{1/2} \cdot \left(\Gamma\left(\frac{n}{2}\right) \right)^{-1} \cdot \int_0^\infty r^{n-1} (\gamma + r^2)^{\frac{1}{1-q}} dr \\ &= 2\pi^{\frac{n}{2}} |\Sigma|^{1/2} \cdot \left(\Gamma\left(\frac{n}{2}\right) \right)^{-1} \cdot \int_0^\infty \gamma^{\frac{n-1}{2} + \frac{1}{1-q}} \left(\frac{r}{\sqrt{\gamma}} \right)^{n-1} \left(1 + \left(\frac{r}{\sqrt{\gamma}} \right)^2 \right)^{\frac{1}{1-q}} dr \\ &= 2\pi^{\frac{n}{2}} |\Sigma|^{1/2} \cdot \left(\Gamma\left(\frac{n}{2}\right) \right)^{-1} \cdot \gamma^{\frac{n}{2} + \frac{1}{1-q}} \cdot \int_0^\infty (r')^{n-1} \left(1 + (r')^2 \right)^{\frac{1}{1-q}} dr' \\ &= 2\pi^{\frac{n}{2}} |\Sigma|^{1/2} \cdot \left(\Gamma\left(\frac{n}{2}\right) \right)^{-1} \cdot \gamma^{\frac{n}{2} + \frac{1}{1-q}} \cdot \int_0^\infty \left((r')^{1-n} \left(1 + (r')^2 \right)^{\frac{1}{q-1}} \right)^{-1} dr' \\ &= 2\pi^{\frac{n}{2}} |\Sigma|^{1/2} \cdot \left(\Gamma\left(\frac{n}{2}\right) \right)^{-1} \cdot \gamma^{\frac{n}{2} + \frac{1}{1-q}} \cdot \frac{1}{2} B\left(\frac{1}{q-1} - \frac{n}{2}, \frac{n}{2}\right) \\ &= \pi^{\frac{n}{2}} |\Sigma|^{1/2} \cdot \left(\Gamma\left(\frac{n}{2}\right) \right)^{-1} \cdot \gamma^{\frac{n}{2} + \frac{1}{1-q}} \cdot \frac{\Gamma\left(\frac{1}{q-1} - \frac{n}{2}\right) \Gamma\left(\frac{n}{2}\right)}{\Gamma\left(\frac{1}{q-1}\right)} \\ &= \pi^{\frac{n}{2}} |\Sigma|^{1/2} \frac{\Gamma\left(\frac{1}{q-1} - \frac{n}{2}\right)}{\Gamma\left(\frac{1}{q-1}\right)} \cdot \gamma^{\frac{n}{2} + \frac{1}{1-q}}. \end{aligned} \quad (23)$$

In step (23), we use the following formula for Beta function [34]:

$$\int_0^\infty \left(x^\alpha (1 + x^\lambda)^\beta \right)^{-1} dx = \frac{1}{\lambda} B\left(\beta - \frac{1-\alpha}{\lambda}, \frac{1-\alpha}{\lambda}\right), \quad (24)$$

where $\alpha < 1$, $\lambda > 0$, $\beta > 0$, $\lambda\beta > 1 - \alpha$. Well-definedness of Z and (23) implies that $\frac{1}{q-1} - \frac{n}{2} > 0$; i.e., $q < 1 + \frac{2}{n}$, which is the reason for the upper bound for the choice of q . (22) implies that

$$\text{tr} \left(\Sigma^{-1} \frac{\int_{\mathbb{R}^n} x x^T \cdot p^q(x) dx}{\int_{\mathbb{R}^n} p^q(x) dx} \right) = \text{tr}(\Sigma^{-1} \Sigma) = n. \quad (25)$$

Hence, since $p(x)$ is symmetric,

$$\begin{aligned} & \text{tr} \left(\Sigma^{-1} \frac{\int_{\mathbb{R}^n} x x^T \cdot p^q(x; q, \Sigma) dx}{\int_{\mathbb{R}^n} p^q(x; q, \Sigma) dx} \right) \\ &= \text{tr} \left(\Sigma^{-1} \frac{\int_{\mathbb{R}_{>0}^n} x x^T \cdot p^q(x; q, \Sigma) dx}{\int_{\mathbb{R}_{>0}^n} p^q(x; q, \Sigma) dx} \right) \\ &= \frac{\text{tr} \left(\int_{\mathbb{R}_{>0}^n} \Sigma^{-1} x x^T \cdot p^q(x; q, \Sigma) dx \right)}{\int_{\mathbb{R}_{>0}^n} p^q(x; q, \Sigma) dx} \\ &= \frac{\int_{\mathbb{R}_{>0}^n} \text{tr}(\Sigma^{-1} x x^T \cdot p^q(x; q, \Sigma)) dx}{\int_{\mathbb{R}_{>0}^n} p^q(x; q, \Sigma) dx} \\ &= \frac{\int_{\mathbb{R}_{>0}^n} \text{tr}(x^T \Sigma^{-1} x) \cdot p^q(x; q, \Sigma) dx}{\int_{\mathbb{R}_{>0}^n} p^q(x; q, \Sigma) dx} \\ &= \frac{\int_{\mathbb{R}_{>0}^n} \langle x, \Sigma^{-1} x \rangle \cdot \left(\frac{1}{Z} (\gamma + \langle x, \Sigma^{-1} x \rangle)^{\frac{1}{1-q}} \right)^q dx}{\int_{\mathbb{R}_{>0}^n} \left(\frac{1}{Z} (\gamma + \langle x, \Sigma^{-1} x \rangle)^{\frac{1}{1-q}} \right)^q dx} \\ &= \frac{\int_{\mathbb{R}_{>0}^n} \langle x, \Sigma^{-1} x \rangle \cdot (\gamma + \langle x, \Sigma^{-1} x \rangle)^{\frac{q}{1-q}} dx}{\int_{\mathbb{R}_{>0}^n} (\gamma + \langle x, \Sigma^{-1} x \rangle)^{\frac{q}{1-q}} dx} \\ &= \frac{\int_{\mathbb{R}_{>0}^n} |\Sigma|^{1/2} \langle x, x \rangle \cdot (\gamma + \langle x, x \rangle)^{\frac{q}{1-q}} dx}{\int_{\mathbb{R}_{>0}^n} |\Sigma|^{1/2} (\gamma + \langle x, x \rangle)^{\frac{q}{1-q}} dx} \\ &= \frac{\int_{\mathbb{R}_{>0}^n} \langle x, x \rangle \cdot (\gamma + \langle x, x \rangle)^{\frac{q}{1-q}} dx}{\int_{\mathbb{R}_{>0}^n} (\gamma + \langle x, x \rangle)^{\frac{q}{1-q}} dx} \\ &= \frac{\int_0^\infty r^{n-1} \cdot r^2 \cdot (\gamma + r^2)^{\frac{q}{1-q}} \cdot \left(\prod_{i=1}^{n-2} \int_0^{\frac{\pi}{2}} \sin^{n-1-i}(\theta_i) d\theta_i \cdot \int_0^{\frac{\pi}{2}} 1 d\theta \right) dr}{\int_0^\infty r^{n-1} \cdot (\gamma + r^2)^{\frac{q}{1-q}} \cdot \left(\prod_{i=1}^{n-2} \int_0^{\frac{\pi}{2}} \sin^{n-1-i}(\theta_i) d\theta_i \cdot \int_0^{\frac{\pi}{2}} 1 d\theta \right) dr} \end{aligned} \quad (26)$$

$$\begin{aligned}
 &= \frac{\int_0^\infty r^{n+1} \cdot (\gamma + r^2)^{\frac{q}{1-q}} dr}{\int_0^\infty r^{n-1} \cdot (\gamma + r^2)^{\frac{q}{1-q}} dr} \\
 &= \frac{\gamma \int_0^\infty \left((r')^{-n-1} \cdot \left(1 + (r')^2 \right)^{\frac{q}{q-1}} \right)^{-1} dr'}{\int_0^\infty \left((r')^{1-n} \cdot \left(1 + (r')^2 \right)^{\frac{q}{q-1}} \right)^{-1} dr'} \\
 &= \frac{\gamma B\left(\frac{q}{q-1} - \frac{n+2}{2}, \frac{n+2}{2}\right)}{B\left(\frac{q}{q-1} - \frac{n}{2}, \frac{n}{2}\right)} \tag{27}
 \end{aligned}$$

$$\begin{aligned}
 &= \gamma \cdot \frac{\Gamma\left(\frac{q}{q-1} - \frac{n}{2} - 1\right) \Gamma\left(\frac{n+2}{2}\right)}{\Gamma\left(\frac{q}{q-1}\right)} / \frac{\Gamma\left(\frac{q}{q-1} - \frac{n}{2}\right) \Gamma\left(\frac{n}{2}\right)}{\Gamma\left(\frac{q}{q-1}\right)} \\
 &= \gamma \cdot \frac{n}{2} \cdot \left(\frac{q}{q-1} - \frac{n}{2} - 1 \right)^{-1}. \tag{28}
 \end{aligned}$$

In step (27), we used (24). Combining (25) and (28), we have

$$\gamma \cdot \frac{n}{2} \cdot \left(\frac{q}{q-1} - \frac{n}{2} - 1 \right)^{-1} = n, \tag{29}$$

which gives that

$$\gamma = \frac{2q}{q-1} - n - 2 = \frac{2}{q-1} - n. \tag{30}$$

Thus, the probability density function that maximizes problem (10) is:

$$\begin{aligned}
 p(x; q, \Sigma) &= \left(\pi^{\frac{n}{2}} |\Sigma|^{1/2} \frac{\Gamma\left(\frac{1}{q-1} - \frac{n}{2}\right)}{\Gamma\left(\frac{1}{q-1}\right)} \cdot \left(\frac{2}{q-1} - n \right)^{\frac{n}{2} + \frac{1}{1-q}} \right)^{-1} \left(\left(\frac{2}{q-1} - n \right) + \langle x, \Sigma^{-1} x \rangle \right)^{\frac{1}{1-q}} \\
 &= \left(\pi^{\frac{n}{2}} |\Sigma|^{1/2} \frac{\Gamma\left(\frac{1}{q-1} - \frac{n}{2}\right)}{\Gamma\left(\frac{1}{q-1}\right)} \cdot \left(\frac{2}{q-1} - n \right)^{\frac{n}{2}} \right)^{-1} \left(1 + \left(\frac{2}{q-1} - n \right)^{-1} \langle x, \Sigma^{-1} x \rangle \right)^{\frac{1}{1-q}} \\
 &= \frac{1}{|\pi \Sigma|^{1/2}} \cdot \frac{\Gamma\left(\frac{1}{q-1}\right)}{\Gamma\left(\frac{1}{q-1} - \frac{n}{2}\right)} \cdot \left(\frac{2}{q-1} - n \right)^{-\frac{n}{2}} \cdot \left(1 + \left(\frac{2}{q-1} - n \right)^{-1} \cdot \langle x, \Sigma^{-1} x \rangle \right)^{\frac{1}{1-q}}.
 \end{aligned}$$

□

When $q \geq 1 + \frac{2}{n}$, the solution to problem (10) does not exist, due to property 1 and discussions in the proof. In Lemma 1, the presented density (20) outlines a formula for

multivariate bell-curve distributions dependent on the value of q . As q shifts from values approaching 1 from above to values approaching $1 + \frac{2}{n}$ from below, the resulting density transitions from Gaussian through a scaled version of the multivariate t -distribution to Cauchy and beyond. This density explicitly details all parameters, enabling the application of the maximum likelihood principle and facilitating the use of maximum likelihood estimation in modeling correlated data performed in Section 5.

In the context of (20), the *characterization matrix* [11], denoted by Σ , can undergo modifications to include the degree of freedom parameter $m := \frac{2}{q-1} - n$ [59]; specifically,

$$p(x; q, \Lambda) = \frac{1}{|\pi\Lambda|^{1/2}} \cdot \frac{\Gamma\left(\frac{m}{2} + \frac{n}{2}\right)}{\Gamma\left(\frac{m}{2}\right)} \cdot (1 + \langle x, \Lambda^{-1}x \rangle)^{\frac{1}{1-q}}, \quad (31)$$

where $\Lambda := m\Sigma$.

$$\text{and } m := \frac{2}{q-1} - n$$

Below are a few useful remarks related to the q Gaussian distribution and other bell-curve distributions.

Remark 2. To incorporate the location parameter μ , (20) and (31) become

$$\begin{aligned} p(x; \mu, q, \Sigma) &= \frac{1}{|\pi\Sigma|^{1/2}} \cdot \frac{\Gamma\left(\frac{1}{q-1}\right)}{\Gamma\left(\frac{1}{q-1} - \frac{n}{2}\right)} \cdot \left(\frac{2}{q-1} - n\right)^{-\frac{n}{2}} \\ &\quad \cdot \left(1 + \left(\frac{2}{q-1} - n\right)^{-1} \cdot \langle x - \mu, \Sigma^{-1}(x - \mu) \rangle\right)^{\frac{1}{1-q}}; \\ p(x; \mu, q, \Lambda) &= \frac{1}{|\pi\Lambda|^{1/2}} \cdot \frac{\Gamma\left(\frac{m}{2} + \frac{n}{2}\right)}{\Gamma\left(\frac{m}{2}\right)} \cdot (1 + \langle x - \mu, \Lambda^{-1}(x - \mu) \rangle)^{\frac{1}{1-q}}. \end{aligned}$$

Remark 3. For random vector $X \sim q\text{Gaussian}(q, \Sigma)$, its q -covariance is

$$\mathbb{E}_q [XX^T] = \left(\frac{1}{q-1} - \frac{n}{2}\right)^{\frac{1-q}{2}} |\pi\Sigma|^{\frac{1-q}{2}} \cdot \frac{\Gamma\left(\frac{q}{q-1} - \frac{n}{2}\right) / \left(\Gamma\left(\frac{1}{q-1} - \frac{n}{2}\right)\right)^q}{\Gamma\left(\frac{q}{q-1}\right) / \left(\Gamma\left(\frac{1}{q-1}\right)\right)^q} \cdot \Sigma. \quad (32)$$

Proof. We note that

$$\begin{aligned} \int_{\mathbb{R}^n} p^q(x; q, \Sigma) dx &= \int_{\mathbb{R}^n} \left(\frac{1}{|\pi\Lambda|^{1/2}} \cdot \frac{\Gamma\left(\frac{1}{q-1}\right)}{\Gamma\left(\frac{1}{q-1} - \frac{n}{2}\right)} \cdot (1 + \langle x, \Lambda^{-1}x \rangle)^{\frac{1}{1-q}} \right)^q dx \\ &= \left(\frac{1}{|\pi\Lambda|^{1/2}} \cdot \frac{\Gamma\left(\frac{1}{q-1}\right)}{\Gamma\left(\frac{1}{q-1} - \frac{n}{2}\right)} \right)^q \cdot \int_{\mathbb{R}^n} \left((1 + \langle x, \Lambda^{-1}x \rangle)^{\frac{1}{1-q}} \right)^q dx \end{aligned}$$

$$\begin{aligned}
 &= \left(\frac{1}{|\pi\Lambda|^{1/2}} \cdot \frac{\Gamma\left(\frac{1}{q-1}\right)}{\Gamma\left(\frac{1}{q-1} - \frac{n}{2}\right)} \right)^q \cdot \int_{\mathbb{R}^n} (1 + \langle x, \Lambda^{-1}x \rangle)^{\frac{q}{1-q}} dx \\
 &= \left(\frac{1}{|\pi\Lambda|^{1/2}} \cdot \frac{\Gamma\left(\frac{1}{q-1}\right)}{\Gamma\left(\frac{1}{q-1} - \frac{n}{2}\right)} \right)^q |\Lambda|^{1/2} \cdot \int_{\mathbb{R}^n} (1 + \langle x, x \rangle)^{\frac{q}{1-q}} dx \\
 &= 2^n \left(\frac{1}{|\pi\Lambda|^{1/2}} \cdot \frac{\Gamma\left(\frac{1}{q-1}\right)}{\Gamma\left(\frac{1}{q-1} - \frac{n}{2}\right)} \right)^q |\Lambda|^{1/2} \cdot \int_{\mathbb{R}_{>0}^n} (1 + \langle x, x \rangle)^{\frac{q}{1-q}} dx \\
 &= 2^n \left(\frac{1}{|\pi\Lambda|^{1/2}} \cdot \frac{\Gamma\left(\frac{1}{q-1}\right)}{\Gamma\left(\frac{1}{q-1} - \frac{n}{2}\right)} \right)^q |\Lambda|^{1/2} \cdot \int_0^\infty r^{n-1} (1 + r^2)^{\frac{q}{1-q}} \\
 &\quad \cdot \left(\prod_{i=1}^{n-2} \int_0^{\frac{\pi}{2}} \sin^{n-1-i}(\theta_i) d\theta_i \cdot \int_0^{\frac{\pi}{2}} 1 d\theta \right) dr \\
 &= \pi 2^{n-1} \left(\frac{1}{|\pi\Lambda|^{1/2}} \cdot \frac{\Gamma\left(\frac{1}{q-1}\right)}{\Gamma\left(\frac{1}{q-1} - \frac{n}{2}\right)} \right)^q |\Lambda|^{1/2} \cdot \left(\prod_{i=1}^{n-2} \frac{1}{2} \frac{\Gamma\left(\frac{n-i}{2}\right) \sqrt{\pi}}{\Gamma\left(\frac{n-i+1}{2}\right)} \right) \\
 &\quad \cdot \int_0^\infty r^{n-1} (1 + r^2)^{\frac{q}{1-q}} dr \\
 &= 2\pi^{\frac{n}{2}} \left(\frac{1}{|\pi\Lambda|^{1/2}} \cdot \frac{\Gamma\left(\frac{1}{q-1}\right)}{\Gamma\left(\frac{1}{q-1} - \frac{n}{2}\right)} \right)^q |\Lambda|^{1/2} \cdot \left(\Gamma\left(\frac{n}{2}\right) \right)^{-1} \cdot \int_0^\infty \left(r^{1-n} (1 + r^2)^{\frac{q}{q-1}} \right)^{-1} dr \\
 &= 2\pi^{\frac{n}{2}} \left(\frac{1}{|\pi\Lambda|^{1/2}} \cdot \frac{\Gamma\left(\frac{1}{q-1}\right)}{\Gamma\left(\frac{1}{q-1} - \frac{n}{2}\right)} \right)^q |\Lambda|^{1/2} \cdot \left(\Gamma\left(\frac{n}{2}\right) \right)^{-1} \cdot \frac{1}{2} B\left(\frac{q}{q-1} - \frac{n}{2}, \frac{n}{2}\right) \\
 &= \pi^{\frac{n}{2}} \left(\frac{1}{|\pi\Lambda|^{1/2}} \cdot \frac{\Gamma\left(\frac{1}{q-1}\right)}{\Gamma\left(\frac{1}{q-1} - \frac{n}{2}\right)} \right)^q |\Lambda|^{1/2} \cdot \left(\Gamma\left(\frac{n}{2}\right) \right)^{-1} \cdot \frac{\Gamma\left(\frac{q}{q-1} - \frac{n}{2}\right) \Gamma\left(\frac{n}{2}\right)}{\Gamma\left(\frac{q}{q-1}\right)} \\
 &= \left(\frac{1}{|\pi\Lambda|^{1/2}} \cdot \frac{\Gamma\left(\frac{1}{q-1}\right)}{\Gamma\left(\frac{1}{q-1} - \frac{n}{2}\right)} \right)^q |\pi\Lambda|^{1/2} \cdot \frac{\Gamma\left(\frac{q}{q-1} - \frac{n}{2}\right)}{\Gamma\left(\frac{q}{q-1}\right)} \\
 &= |\pi\Lambda|^{\frac{1-q}{2}} \cdot \frac{\Gamma\left(\frac{q}{q-1} - \frac{n}{2}\right) / \left(\Gamma\left(\frac{1}{q-1} - \frac{n}{2}\right) \right)^q}{\Gamma\left(\frac{q}{q-1}\right) / \left(\Gamma\left(\frac{1}{q-1}\right) \right)^q}.
 \end{aligned}$$

From (13), we then have the following expression for the q -variance-covariance matrix

$$\begin{aligned}
 \mathbb{E}_q [XX^T] &= \int_{\mathbb{R}^n} xx^T \cdot p^q(x; q, \Sigma) dx \\
 &= \int_{\mathbb{R}^n} p^q(x; q, \Sigma) dx \cdot \Sigma \\
 &= |\pi\Lambda|^{\frac{1-q}{2}} \cdot \frac{\Gamma\left(\frac{q}{q-1} - \frac{n}{2}\right) / \left(\Gamma\left(\frac{1}{q-1} - \frac{n}{2}\right)\right)^q}{\Gamma\left(\frac{q}{q-1}\right) / \left(\Gamma\left(\frac{1}{q-1}\right)\right)^q} \cdot \Sigma \\
 &= m^{\frac{1-q}{2}} |\pi\Sigma|^{\frac{1-q}{2}} \cdot \frac{\Gamma\left(\frac{q}{q-1} - \frac{n}{2}\right) / \left(\Gamma\left(\frac{1}{q-1} - \frac{n}{2}\right)\right)^q}{\Gamma\left(\frac{q}{q-1}\right) / \left(\Gamma\left(\frac{1}{q-1}\right)\right)^q} \cdot \Sigma \\
 &= \left(\frac{1}{q-1} - \frac{n}{2}\right)^{\frac{1-q}{2}} |\pi\Sigma|^{\frac{1-q}{2}} \Sigma \cdot \frac{\Gamma\left(\frac{q}{q-1} - \frac{n}{2}\right) / \left(\Gamma\left(\frac{1}{q-1} - \frac{n}{2}\right)\right)^q}{\Gamma\left(\frac{q}{q-1}\right) / \left(\Gamma\left(\frac{1}{q-1}\right)\right)^q}. \quad (33)
 \end{aligned}$$

□

Remark 4. Multivariate t distributions with degree of freedom of $\frac{2}{q-1} - n$ and the scale matrix of Σ are q Gaussian with shape parameter q and scale matrix Σ .

Remark 5. The multivariate Cauchy distributions [31] in $\mathbb{R}^n$ with scale matrix $\frac{1}{2}\Sigma$ are q Gaussian with shape parameter $q = 1 + \frac{2}{n+1}$ and scale matrix Σ .

Remark 6. For a random vector $X \sim q\text{Gaussian}(q, \Sigma)$, the variance-covariance matrix exists if and only if $q < 1 + \frac{2}{n+2}$; following the same procedure to derive the variance-covariance matrix for multivariate t distribution yields that, *if existing*,

$$\mathbb{E} [XX^T] = \frac{m}{m-2} \Sigma, \quad (34)$$

where $m = \frac{2}{q-1} - n$.

The remarks above on q Gaussian distributions reveal their flexibility in incorporating a location parameter, μ , and adapting to multivariate contexts through detailed formulas. Remarkably, these distributions bridge with the class of multivariate bell curve distributions including Gaussian, scaled t , and Cauchy distributions under certain conditions on the shape parameter q . The elaboration of q -correlation and the q -variable-covariance matrix underscores the capability of these distributions to model and understand the intricacies of correlated data effectively.

4 Proximal Conjugate Gradient Algorithm

As delineated in Section 5.3, our optimization scenario is predominantly quadratic in nature; therefore, the conjugate gradient approach has potential for fast convergence and

numerical stability. This insight forms the basis for our introduction of a proximal conjugate gradient algorithm framework, tailored to navigate the complexities introduced by the nonconvex penalized q Gaussian likelihood function for sparse statistical learning. To lay the groundwork for this discussion, we begin with an overview of relevant concepts in variational and nonsmooth analysis, presented in Section 4.1. The results presented in Section 4.1 can be found in recent textbooks on variational and nonsmooth analysis, such as [46, 10, 35, 36, 37, 3].

4.1 A Review on Variational and Nonsmooth Analysis

Let $\mathcal{C}^{k,\alpha_H}$ with $k \in \mathbb{N}_{\geq 0}$ and $\alpha_H \in [0, 1]$ denote the function space such that $\forall F \in \mathcal{C}^{k,\alpha_H}$, F is k th continuously differentiable, and $D^k F$ is globally Hölder continuous with exponent α_H ; clearly, when $\alpha_H = 1$, $D^k F$ is globally Lipschitz continuous. In this subsection, we will state the results from variational and nonsmooth analysis related to the following optimization problem:

$$\min_{x \in \mathbb{R}^{p+1}} f(x) := g(x) + h(x), \quad (35)$$

where $f \in \mathcal{C}^{0,0}(\mathbb{R}^{p+1}, \mathbb{R})$ is a locally-Lipschitz proper function, $g \in \mathcal{C}^{1,1}(\mathbb{R}^{p+1}, \mathbb{R})$ is globally $L_{\nabla g}$ -smooth and possibly nonconvex, and $h \in \mathcal{C}^{0,0}(\mathbb{R}^{p+1}, \mathbb{R})$ is a convex locally-Lipschitz function, possibly nonsmooth. The globally Lipschitz property of ∇g can be alternatively addressed by carrying out the optimization over a compact set. In such scenarios, given that ∇g is locally Lipschitz, it inherently becomes globally Lipschitz when restricted to a compact set.

Results from convex analysis suggest that g, h are Clarke regular; thus, f is Clarke regular. The Clarke's directional derivative, defined by

$$\begin{aligned} f^\circ(x; d) &:= \limsup_{y \rightarrow x, t \searrow 0} \frac{f(y + td) - f(y)}{t} \\ &= \inf_{\delta > 0} \sup_{\|y - x\| \leq \delta, 0 < t < \delta} \frac{f(y + td) - f(y)}{t}, \end{aligned}$$

exists for all $x \in \mathbb{R}^{p+1}$ since f is Clarke regular. The Clarke subdifferential, denoted by $\partial_\circ$, is a set-valued mapping defined by

$$\partial_\circ f(x) := \{\phi \in \mathbb{R}^{p+1} \mid \forall d \in \mathbb{R}^{p+1}, \langle \phi, d \rangle \leq f^\circ(x; d)\}. \quad (36)$$

Since f is a locally Lipschitz function, $\forall x \in \mathbb{R}^{p+1}$, $\partial_\circ f(x) \neq \emptyset$. Fundamental convex analysis results show that $\forall x \in \mathbb{R}^{p+1}$, $\partial_\circ f(x)$ is compact, convex, and upper-semicontinuous. $\forall x, d \in \mathbb{R}^{p+1}$, and we also have

$$f^\circ(x; d) = \max_{u \in \partial_\circ f(x)} \left\langle u, \frac{d}{\|d\|} \right\rangle. \quad (37)$$

Furthermore, (37) is upper-semicontinuous with respect to x . Simple convex geometry results conclude that

$$\{(v, -1) \mid v \in \partial_{\circ} f(x)\} = N_{\text{epi } f}(x, f(x)), \quad (38)$$

where $N_{\text{epi } f}(x, f(x))$ denotes the *normal cone* to $\text{epi } f$ at the point $(x, f(x))$.

Since g is smooth, $\partial_{\circ} g(x) = \{\nabla g(x)\}$ is a singleton. Then $\partial_{\circ} f(x) = \partial_{\circ} g(x) + \partial_{\circ} h(x)$, and

$$\begin{aligned} f^{\circ}(x; d) &= \max_{u \in \partial_{\circ} f(x)} \left\langle u, \frac{d}{\|d\|} \right\rangle \\ &= \max_{u \in (\nabla g(x) + \partial_{\circ} h(x))} \left\langle u, \frac{d}{\|d\|} \right\rangle \\ &= \left\langle \nabla g(x), \frac{d}{\|d\|} \right\rangle + \max_{v \in \partial_{\circ} h(x)} \left\langle v, \frac{d}{\|d\|} \right\rangle \\ &= g^{\circ}(x; d) + h^{\circ}(x; d). \end{aligned} \quad (39)$$

Let

$$M_{\rho} t(x) := \left(t \square \left(\frac{1}{2\rho} \|\cdot\|^2 \right) \right)(x) = \inf_{y \in \mathbb{R}^{p+1}} t(y) + \frac{1}{2\rho} \|y - x\|^2 \quad (40)$$

denote the Moreau envelope operator parameterized by $\rho \in \mathbb{R}_{>0}$ applied on an arbitrary proper, lower semi-continuous, locally Lipschitz function $t \in \mathcal{C}^{0,0}(\mathbb{R}^{p+1}, \mathbb{R})$, where “ $\square$ ” denotes the infimal convolution operator. We have that the Moreau envelope is a smoothing operator, specifically,

$$\text{epi } t + \text{epi } \frac{1}{2\rho} \|\cdot\|^2 \subseteq \text{epi } M_{\rho} t, \quad (41)$$

where “epi” denotes the epigraph. Clearly, $M_{\rho} t(x) \leq t(x)$, since $(0, 0) \in \text{epi } \frac{1}{2\rho} \|\cdot\|^2$ implies that $\text{epi } t = \text{epi } t + (0, 0) \subseteq \text{epi } t + \text{epi } \frac{1}{2\rho} \|\cdot\|^2 \subseteq \text{epi } M_{\rho} t$. When t is convex, (41) takes the equal sign; i.e., the infimal convolution becomes the exact infimal convolution.

Consider the affine function

$$A(x) := \langle a, x \rangle + b, \quad (42)$$

simple algebra shows that the Moreau envelope applied on A is

$$M_{\rho} A(x) = \langle a, x \rangle + b + \frac{\rho}{2} \|a\|^2 = A(x) + \frac{\rho}{2} \|a\|^2 \quad (43)$$

for some $a, b \in \mathbb{R}^{p+1}$. Moreover, the following affine addition property is often used in proximal algorithms, mainly due to the fact that the epigraph of an affine function is a half-space that the Moreau envelope applied on:

$$M_{\rho}(t + A)(x) = M_{\rho} t(x - \rho a) + \langle a, x \rangle + b - \frac{\rho}{2} \|a\|^2 \quad (44)$$

Let

$$\text{prox}_{\rho t}(x) := \arg M_{\rho t}(x) = \arg \min_{y \in \mathbb{R}^{p+1}} t(y) + \frac{1}{2\rho} \|y - x\|^2 \quad (45)$$

denote the proximal operator, a set-valued mapping; we have

$$\text{prox}_{\rho t} = (I + \rho \partial_{\circ} t)^{-1} \quad (46)$$

is the resolvent of the Clarke's subdifferential operator $\rho \partial_{\circ} t$.

For nonsmooth problems, proximal methods are often used. Fundamental convex analysis results show that:

1. the Moreau envelope $M_{\rho t}(x)$ is twice differentiable; thus, its gradient $\nabla M_{\rho t}(x)$ is well-defined.
2. If t is convex, $\text{prox}_{\rho t}(x)$ is a singleton. *For the sake of parsimony, with a slight abuse of notation, we use $\text{prox}_{\rho t}$ to represent a function in this case.* It follows that both $\text{prox}_{\rho t}$ and $\nabla M_{\rho t}$ are firmly non-expansive, and Moreau's decomposition theorem implies that

$$\nabla M_{\rho t}(x) = \rho^{-1} (x - \text{prox}_{\rho t}(x)). \quad (47)$$

The results from variational and nonsmooth analysis in this subsection have laid the foundation for proving the properties discussed in Section 4.2.

4.2 Proximal Conjugate Gradient Framework

Proximal methods are powerful optimization techniques and are particularly adept at handling problems characterized by sparsity, which usually leads to an optimization problem that is nonsmooth [40]. Proximal algorithm tends to outperform other methods by far for nonsmooth problems [64, 32]. On another ground, Krylov subspace methods represent a cornerstone of numerical analysis, providing a powerful framework for solving large-scale optimization problems efficiently [49]. Krylov subspace methods exhibit a remarkable property of convergence acceleration and vastly improved numerical stability, making them indispensable tools in the numerical analyst's toolkit.

Having reviewed the related results from variational and non-smooth analysis in Section 4.1, we are ready to introduce our main optimization framework to combine proximal methods and conjugate gradient together. The essence of proximal algorithms lies upon the Moreau envelope's smoothing on the objective function. Indeed, proximal methods minimize $M_{\rho}f$ instead of f , thus avoiding nonsmoothness since $M_{\rho}f$ is a smooth function. In this view, proximal algorithms are, in fact, minimizing the Moreau envelope of the objective function. Thus, a wide class of numerical optimization algorithms can easily have their proximal version. Among those, conjugate gradients, a type of Krylov subspace method, are the state-of-the-art methods in smooth optimization due to their computational and memory efficiency, scalability, and numerical stability.

Prior to introducing our proximal conjugate gradient update framework, we will first show the equivalency of the optimization problem to minimize (35) and its the Moreau envelope. In nonconvex optimization, the main task for numerical optimization is to find a Clarke stationary point of the objective function, for which we show in Theorem 7 that the set of Clarke stationary point of f is identical to that of $M_\rho f$ for $\rho \in (0, L_{\nabla g}^{-1})$.

Lemma 7. $\forall \bar{x} \in \mathbb{R}^{p+1}, \rho \in (0, L_{\nabla g}^{-1}),$

$$0 \in \partial_\circ f(\bar{x}) \Leftrightarrow \nabla M_\rho f(\bar{x}) = 0, \quad (48)$$

Proof. Consider arbitrary $x \in \mathbb{R}^{p+1}, \rho \in (0, L_{\nabla g}^{-1})$. As discussed previously, the gradient of the Moreau envelope $\nabla M_\rho f(x) = \rho^{-1}(x - \text{prox}_{\rho f}(x))$ implies that

$$\text{prox}_{\rho f}(x) = x - \rho \nabla M_\rho f(x), \quad (49)$$

which implies the following first-order (necessary) optimality condition for Clarke's stationary point:

$$0 \in \rho^{-1}(x - \rho \nabla M_\rho f(x) - x) + \partial_\circ f(x - \rho \nabla M_\rho f(x)). \quad (50)$$

The relation above is simplified to

$$\nabla M_\rho f(x) \in \partial_\circ f(x - \rho \nabla M_\rho f(x)) = \nabla g(x - \rho \nabla M_\rho f(x)) + \partial_\circ h(x - \rho \nabla M_\rho f(x)). \quad (51)$$

Consider arbitrary $\bar{x} \in \mathbb{R}^{p+1}, \rho \in (0, L_{\nabla g}^{-1})$.

“ $\Rightarrow$ ” of (48):

Let $0 \in \partial_\circ f(\bar{x}) = \nabla g(\bar{x}) + \partial_\circ h(\bar{x})$; i.e., $\bar{x}$ is a Clarke stationary point of f . Then $-\nabla g(\bar{x}) \in \partial_\circ h(\bar{x})$. Since h is convex, (51) implies that

$$\langle -\nabla g(\bar{x}) - (\nabla M_\rho f(\bar{x}) - \nabla g(\bar{x} - \rho \nabla M_\rho f(\bar{x}))), \rho \nabla M_\rho f(\bar{x}) \rangle \geq 0. \quad (52)$$

Simplification gives

$$\langle \nabla g(\bar{x} - \rho \nabla M_\rho f(\bar{x})) - \nabla g(\bar{x}), \nabla M_\rho f(\bar{x}) \rangle \geq \|\nabla M_\rho f(\bar{x})\|^2. \quad (53)$$

By Cauchy-Schwartz inequality,

$$\langle \nabla g(\bar{x} - \rho \nabla M_\rho f(\bar{x})) - \nabla g(\bar{x}), \nabla M_\rho f(\bar{x}) \rangle \leq L_{\nabla g} \cdot \rho \|\nabla M_\rho f(\bar{x})\|^2. \quad (54)$$

Since $\rho < L_{\nabla g}^{-1}$, (53) and (54) imply that

$$\|\nabla M_\rho f(\bar{x})\|^2 \leq \langle \nabla g(\bar{x} - \rho \nabla M_\rho f(\bar{x})) - \nabla g(\bar{x}), \nabla M_\rho f(\bar{x}) \rangle < \|\nabla M_\rho f(\bar{x})\|^2, \quad (55)$$

which implies that

$$\nabla M_\rho f(\bar{x}) = 0; \quad (56)$$

i.e., $\bar{x}$ is the stationary point of $M_\rho f$, hence a Clarke's stationary point.

“ $\Leftarrow$ ” of (48):

Let $\nabla f_\rho(\bar{x}) = 0$; i.e. $\bar{x}$ is a stationary point of $M_\rho f$. It follows directly from (51) that

$$0 = \nabla M_\rho f(\bar{x}) \in \partial_\circ f(\bar{x} - \rho \nabla M_\rho f(\bar{x})) = \partial_\circ f(\bar{x}); \quad (57)$$

i.e., $\bar{x}$ is a Clarke stationary point of f . $\square$

The vast majority of optimization algorithms for smooth objective functions require Lipschitz continuity of the objective function. Thus, we are to propose the following Lemma to show the Lipschitz continuity of the gradient of the Moreau envelope of f .

Lemma 8. $\forall \rho \in (0, L_{\nabla g}^{-1}), \exists L_{\nabla M_\rho f} \in \mathbb{R}_{>0}$ such that

$$\forall x, y \in \mathbb{R}^{p+1}, \|\nabla M_\rho f(x) - \nabla M_\rho f(y)\| \leq L_{\nabla M_\rho f} \|x - y\|. \quad (58)$$

Proof. Consider arbitrary $x, y \in \mathbb{R}^{p+1}$. From (51), since h is convex,

$$\begin{aligned} & \langle \nabla M_\rho f(x) - \nabla g(x - \rho \nabla M_\rho f(x)) - (\nabla M_\rho f(y) - \nabla g(y - \rho \nabla M_\rho f(y))), x \\ & - \rho \nabla M_\rho f(x) - (y - \rho \nabla M_\rho f(y)) \rangle \geq 0. \end{aligned} \quad (59)$$

Simplification gives

$$\begin{aligned} & \langle \nabla M_\rho f(x) - \nabla M_\rho f(y) - (\nabla g(x - \rho \nabla M_\rho f(x)) - \nabla g(y - \rho \nabla M_\rho f(y))), x \\ & - y - \rho(\nabla M_\rho f(x) - \nabla M_\rho f(y)) \rangle \geq 0. \end{aligned} \quad (60)$$

Let $\delta_{\nabla M_\rho f} := \nabla M_\rho f(x) - \nabla M_\rho f(y)$, $\delta_{\nabla g} := \nabla g(x - \rho \nabla f_\rho(x)) - \nabla g(y - \rho \nabla f_\rho(y))$, and $\delta_{x,y} := x - y$, then

$$\begin{aligned} 0 & \leq \langle \delta_{\nabla M_\rho f} - \delta_{\nabla g}, \delta_{x,y} - \rho \delta_{\nabla M_\rho f} \rangle \\ & = -\rho \|\delta_{\nabla M_\rho f}\|^2 + \rho \langle \delta_{\nabla g}, \delta_{\nabla M_\rho f} \rangle + \langle \delta_{\nabla M_\rho f}, \delta_{x,y} \rangle - \langle \delta_{\nabla g}, \delta_{x,y} \rangle \\ & \leq -\rho \|\delta_{\nabla M_\rho f}\|^2 + \rho \|\delta_{\nabla g}\| \cdot \|\delta_{\nabla M_\rho f}\| + \|\delta_{\nabla M_\rho f}\| \cdot \|\delta_{x,y}\| + \|\delta_{\nabla g}\| \cdot \|\delta_{x,y}\| \\ & \leq -\rho \|\delta_{\nabla M_\rho f}\|^2 + \rho L_{\nabla g} (\|\delta_{x,y}\| + \rho \|\delta_{\nabla M_\rho f}\|) \cdot \|\delta_{\nabla M_\rho f}\| \\ & \quad + \|\delta_{\nabla M_\rho f}\| \cdot \|\delta_{x,y}\| + L_{\nabla g} (\|\delta_{x,y}\| + \rho \|\delta_{\nabla M_\rho f}\|) \cdot \|\delta_{x,y}\| \end{aligned}$$

Simplification of the above inequality gives

$$\|\delta_{\nabla M_\rho f}\| \leq \frac{2L_g \rho + 1 + \sqrt{8L_g \rho + 1}}{2\rho(1 - L_{\nabla g} \rho)} \|\delta_{x,y}\|; \quad (61)$$

i.e.,

$$\|\nabla M_\rho f(x) - \nabla M_\rho f(y)\| \leq L_{\nabla M_\rho f} \|x - y\|, \quad (62)$$

where

$$L_{\nabla M_\rho f} := \frac{2L_g\rho + 1 + \sqrt{8L_g\rho + 1}}{2\rho(1 - L_{\nabla g}\rho)} > 0. \quad (63)$$

Following this idea, we introduce our proximal conjugate gradient framework in Algorithm 1. $\square$

Algorithm 1 Proximal Point Algorithm

- 1: Input: A fixed value of $\rho \in (0, \rho^{-1})$
 - 2: Calculate the gradient of the Moreau envelope: $s^{(k)} := \nabla M_\rho f(x^{(k)})$
 - 3: $d^{(k)} := -s^{(k)} + \beta^{(k)} \cdot d^{(k-1)}$
 - 4: Line search to find $\alpha^{(k)}$ for the update $x^{(k+1)} := x^{(k)} + \alpha^{(k)} d^{(k)}$
 - 5: Update $x^{(k+1)} := x^{(k)} + \alpha^{(k)} d^{(k)}$
-

In the above algorithm, $\beta^{(k)}$ is the conjugate parameter. The significant meaning of Algorithm 1 is that for any global convergent numerical method to find the equilibria of a globally Lipschitz flow, which generally include the global convergent first-order methods, Algorithm 1 can transform such a method to a proximal counterpart.

For some objective functions, the gradient of the Moreau envelope can be calculated directly. However, calculation for the Moreau envelope's gradient is not tractable for many objective functions whose smooth component g is of complicated form. Motivated by this, we further consider the following the Moreau envelope of the objective function with linearized g , such linearization step is frequently used in proximal algorithms for statistical sparse learning problems (e.g., [38, 20, 63]).

Consider the linearized surrogate of (35), the locally Lipschitz function $\tilde{f} \in \mathcal{C}^{0,0}(\mathbb{R}^{p+1}, \mathbb{R})$, defined by

$$\tilde{f}(x; u) := \langle u, x \rangle + h(x) \quad (64)$$

$$\text{prox}_{\rho\tilde{f}}(x; u) = \arg \min_{y \in \mathbb{R}^{p+1}} \left\{ \langle u, y \rangle + \frac{1}{2\rho} \|y - x\|^2 + h(y) \right\} \quad (65)$$

$$\nabla_x M_\rho \tilde{f}(x; u) = \rho^{-1} \left(x - \text{prox}_{\rho\tilde{f}}(x; u) \right) \quad (66)$$

$\text{prox}_{\rho\tilde{f}}(x; u)$ is the proximal operator applied on $\tilde{f}$, and $\nabla_x M_\rho \tilde{f}(x; u)$ is the gradient of the Moreau envelope of $\tilde{f}$. The linearization term $\langle u, x \rangle$ in (64) depends on u . Recognize that $\tilde{f}(x; u)$ is linearizing the nonconvex smooth component g in (101) when $u = \nabla g(x)$.

We establish several definitions for subsequent utilization. Define the mapping $\tilde{g}_\rho = I - \rho \nabla g \in \mathcal{C}^{0,0}(\mathbb{R}^{p+1}, \mathbb{R}^{p+1})$ for some $\rho \in (0, L_{\nabla g}^{-1})$, the locally Lipschitz property of $\tilde{g}_\rho$

follows from $g \in \mathcal{C}^{1,1}$; i.e., $\tilde{g}_\rho(x) := x - \rho \nabla g(x)$. The following Lemma identifies some fundamental property of $\tilde{g}_\rho$.

Lemma 9. $\tilde{g}_\rho$ is a bijective from $\mathbb{R}^{p+1}$ to $\mathbb{R}^{p+1}$, and $\tilde{g}_\rho^{-1}$ is globally Lipschitz with constant $(1 - \rho L_{\nabla g})^{-1}$.

Proof. Injectivity proof:

Consider arbitrary $x_1, x_2 \in \mathbb{R}^{p+1}$. Since $\rho \in (0, L_{\nabla g}^{-1})$, $x_1 - \rho \nabla g(x_1) = x_2 - \rho \nabla g(x_2)$ implies that

$$\|x_1 - x_2\| = \rho \|\nabla g(x_1) - \nabla g(x_2)\| \leq \rho L_{\nabla g} \|x_1 - x_2\| < \|x_1 - x_2\|, \quad (67)$$

hence $x_1 = x_2$. This shows that $\tilde{g}_\rho$ is an injective mapping.

Surjectivity proof:

Consider arbitrary $y_1, y_2 \in \mathbb{R}^{p+1}$. Consider arbitrary $z \in \mathbb{R}^{p+1}$. Define mapping $\mathcal{T}(y) := z + \rho \nabla g(y)$, then

$$\begin{aligned} \|\mathcal{T}(y_1) - \mathcal{T}(y_2)\| &= \|z + \rho \nabla g(y_1) - (z + \rho \nabla g(y_2))\| \\ &= \rho \|\nabla g(y_1) - \nabla g(y_2)\| \\ &\leq \rho L_{\nabla g} \|y_1 - y_2\| \\ &< \|y_1 - y_2\|. \end{aligned}$$

Thus, $\mathcal{T}$ is a contraction mapping, since $\mathbb{R}^{p+1}$ equipped with Euclidean topology is a Banach space, by Banach fixed point theorem, $\mathcal{T}$ has a fixed point; i.e., $\exists y \in \mathbb{R}^{p+1}$ such that $y = z + \rho \nabla g(y)$, or equivalently, $\tilde{g}_\rho(y) = y - \rho \nabla g(y) = z$. Thus, $\mathbb{R}^{p+1} \subseteq \tilde{g}_\rho(\mathbb{R}^{p+1})$.

Globally Lipschitz constant derivation for inverse map:

Since ∇g is globally $L_{\nabla g}$ -Lipschitz,

$$\begin{aligned} \|\tilde{g}_\rho(y_1) - \tilde{g}_\rho(y_2)\| &= \|y_1 - \rho \nabla g(y_1) - (y_2 - \rho \nabla g(y_2))\| \\ &= \|y_1 - y_2 - \rho(\nabla g(y_1) - \nabla g(y_2))\| \\ &\geq \| \|y_1 - y_2\| - \rho \|\nabla g(y_1) - \nabla g(y_2)\| \| \\ &= \| \|y_1 - y_2\| - \rho \|\nabla g(y_1) - \nabla g(y_2)\| \| \\ &= \|y_1 - y_2\| - \rho \|\nabla g(y_1) - \nabla g(y_2)\| \\ &\geq (1 - \rho L_{\nabla g}) \|y_1 - y_2\| \end{aligned} \quad (68)$$

where (68) is due to the fact that

$$\rho \|\nabla g(y_1) - \nabla g(y_2)\| \leq \rho L_{\nabla g} \|y_1 - y_2\| < \|y_1 - y_2\|. \quad (69)$$

Since $\tilde{g}_\rho$ is surjective, consider arbitrary $z_1, z_2 \in \mathbb{R}^{p+1}$ let $y_1 := \tilde{g}_\rho^{-1}(z_1)$ and $y_2 := \tilde{g}_\rho^{-1}(z_2)$, then

$$\|\tilde{g}_\rho^{-1}(z_1) - \tilde{g}_\rho^{-1}(z_2)\| \leq (1 - \rho L_{\nabla g})^{-1} \|z_1 - z_2\|. \quad (70)$$

□

Define

$$\mathcal{G}_{\rho\tilde{f}}(x) := \nabla_x M_{\rho\tilde{f}}(x; u) \quad (71)$$

with $u = \nabla g(x)$; i.e., $\mathcal{G}_{\rho\tilde{f}}(x)$ is the gradient of the Moreau envelope of $\tilde{f}$.

Similarly to Lemma 7 and 8, we are to prove that the set of Clarke's stationary of (101) is identical to the set $\{\bar{x} \in \mathbb{R}^{p+1} | \mathcal{G}_{\rho\tilde{f}}(\bar{x}) = 0\}$ in Lemma 10, and then we are to show that (71) is globally Lipschitz in Lemma 11.

Lemma 10. $\forall \bar{x} \in \mathbb{R}^{p+1}, \rho \in \mathbb{R}_{>0}$,

$$0 \in \partial_{\circ} f(\bar{x}) \Leftrightarrow \mathcal{G}_{\rho\tilde{f}}(\bar{x}) = 0. \quad (72)$$

Proof. Consider arbitrary $x \in \mathbb{R}^{p+1}$. The $\tilde{f}$ is convex since it is a sum of convex function h and a linear mapping of x , which is convex.

$$\mathcal{G}_{\rho\tilde{f}}(x) = \rho^{-1} \left(x - \text{prox}_{\rho\tilde{f}}(x; \nabla g(x)) \right) \quad (73)$$

$$= \rho^{-1} \left(x - \text{prox}_{\rho h}(x - \rho \nabla g(x)) \right) \quad (74)$$

$$\begin{aligned} &= \nabla g(x) + \rho^{-1} \left(x - \rho \nabla g(x) - \text{prox}_{\rho h}(x - \rho \nabla g(x)) \right) \\ &= \nabla g(x) + (\nabla M_{\rho} h) \circ \tilde{g}_{\rho}(x) \end{aligned} \quad (75)$$

(74) is due to the affine addition property of proximal mapping. From (64) and (73),

$$\begin{aligned} &\mathcal{G}_{\rho\tilde{f}}(x) = \rho^{-1} \left(x - \text{prox}_{\rho\tilde{f}}(x, \nabla g(x)) \right) \\ \Rightarrow &\text{prox}_{\rho\tilde{f}}(x, \nabla g(x)) = x - \rho \cdot \mathcal{G}_{\rho\tilde{f}}(x) \\ \Rightarrow &0 \in \rho^{-1} \left(x - \rho \cdot \mathcal{G}_{\rho\tilde{f}}(x) - x \right) + \partial_{\circ} \tilde{f} \left(x - \rho \cdot \mathcal{G}_{\rho\tilde{f}}(x) \right) \\ \Rightarrow &0 \in -\mathcal{G}_{\rho\tilde{f}}(x) + \nabla g(x) + \partial_{\circ} h \left(x - \rho \cdot \mathcal{G}_{\rho\tilde{f}}(x) \right) \\ \Rightarrow &\mathcal{G}_{\rho\tilde{f}}(x) \in \nabla g(x) + \partial_{\circ} h \left(x - \rho \cdot \mathcal{G}_{\rho\tilde{f}}(x) \right) \\ \Rightarrow &\mathcal{G}_{\rho\tilde{f}}(x) - \nabla g(x) \in \partial_{\circ} h \left(x - \rho \cdot \mathcal{G}_{\rho\tilde{f}}(x) \right). \end{aligned} \quad (76)$$

Thus, since h is convex, $\forall v \in \partial_{\circ} h(x)$,

$$\begin{aligned} &\left\langle \mathcal{G}_{\rho\tilde{f}}(x) - \nabla g(x) - v, x - \rho \cdot \mathcal{G}_{\rho\tilde{f}}(x) - x \right\rangle \geq 0 \\ \Rightarrow &\left\langle \mathcal{G}_{\rho\tilde{f}}(x) - \nabla g(x) - v, \mathcal{G}_{\rho\tilde{f}}(x) \right\rangle \leq 0 \\ \Rightarrow &\left\| \mathcal{G}_{\rho\tilde{f}}(x) \right\|^2 \leq \left\langle \nabla g(x) + v, \mathcal{G}_{\rho\tilde{f}}(x) \right\rangle \\ &\leq \|\nabla g(x) + v\| \cdot \left\| \mathcal{G}_{\rho\tilde{f}}(x) \right\| \end{aligned} \quad (77)$$

$$\implies \left\| \mathcal{G}_{\rho\tilde{f}}(x) \right\| \leq \|\nabla g(x) + v\|, \quad (78)$$

provided that $\left\| \mathcal{G}_{\rho\tilde{f}}(x) \right\| \neq 0$. Basic results on the Moreau envelope shows that $\mathcal{G}_{\rho\tilde{f}}(x) = 0$ implies that x is a Clarke stationary point of $\tilde{f}(x)$.

Now we are proceed to prove (72):

“ $\implies$ ”:

Consider arbitrary $\bar{x} \in \mathbb{R}^{p+1}$ and $\rho \in \mathbb{R}_{>0}$. Let $0 \in \partial_{\circ} f(\bar{x}) = \nabla g(\bar{x}) + \partial_{\circ} h(\bar{x})$; i.e., $\bar{x}$ is a Clarke stationary point of f . Then $\exists v \in \partial_{\circ} h(\bar{x})$ such that $\nabla g(\bar{x}) + v = 0$. (78) implies that

$$\left\| \mathcal{G}_{\rho\tilde{f}}(\bar{x}) \right\| \leq \|\nabla g(\bar{x}) + v\| = 0. \quad (79)$$

Thus, $\mathcal{G}_{\rho\tilde{f}}(\bar{x}) = 0$.

“ $\Leftarrow$ ”:

Consider arbitrary $\bar{x} \in \mathbb{R}^{p+1}$. Let $\mathcal{G}_{\rho\tilde{f}}(\bar{x}) = 0$; i.e., $\mathcal{G}_{\rho\tilde{f}}(\bar{x}) = 0$ is stationary. (76) implies that

$$0 = \mathcal{G}_{\rho\tilde{f}}(\bar{x}) \in \nabla g(\bar{x}) + \partial_{\circ} h\left(\bar{x} - \rho \cdot \mathcal{G}_{\rho\tilde{f}}(\bar{x})\right) = \nabla g(\bar{x}) + \partial_{\circ} h(\bar{x}) = \partial_{\circ} f(\bar{x}). \quad (80)$$

Thus, $\bar{x}$ is a Clarke stationary point of f . $\square$

Lemma 11. $\forall \rho \in \mathbb{R}_{>0}, \exists L_{\mathcal{G}_{\rho\tilde{f}}} \in \mathbb{R}_{>0}$ such that

$$\forall x, y \in \mathbb{R}^{p+1}, \left\| \mathcal{G}_{\rho\tilde{f}}(x) - \mathcal{G}_{\rho\tilde{f}}(y) \right\| \leq L_{\mathcal{G}_{\rho\tilde{f}}} \|x - y\|. \quad (81)$$

Proof. Consider arbitrary $x, y \in \mathbb{R}^{p+1}$ and $\rho \in \mathbb{R}_{>0}$. Let $u := \nabla g(x)$ and $v := \nabla g(y)$,

$$\begin{aligned} \left\| \mathcal{G}_{\rho\tilde{f}}(x) - \mathcal{G}_{\rho\tilde{f}}(y) \right\| &= \left\| \nabla_x M_{\rho\tilde{f}}(x; u) - \nabla_y M_{\rho\tilde{f}}(y; v) \right\| \\ &= \left\| \nabla_x M_{\rho\tilde{f}}(x; u) - \nabla_x M_{\rho\tilde{f}}(x; v) + \nabla_x M_{\rho\tilde{f}}(x; v) - \nabla_y M_{\rho\tilde{f}}(y; v) \right\| \\ &\leq \left\| \nabla_x M_{\rho\tilde{f}}(x; u) - \nabla_x M_{\rho\tilde{f}}(x; v) \right\| + \left\| \nabla_x M_{\rho\tilde{f}}(x; v) - \nabla_y M_{\rho\tilde{f}}(y; v) \right\| \\ &\leq \|u - v\| + \left\| \nabla_x M_{\rho\tilde{f}}(x; v) - \nabla_y M_{\rho\tilde{f}}(y; v) \right\| \end{aligned} \quad (82)$$

$$\begin{aligned} &\leq \|u - v\| + \rho^{-1} \|x - y\| \\ &\leq L_{\nabla g} \|x - y\| + \rho^{-1} \|x - y\| \\ &= (L_{\nabla g} + \rho^{-1}) \|x - y\| \end{aligned} \quad (83)$$

(82) is due to Lemma 4 in [20], and (83) is due to $\tilde{f}(\cdot; v)$ is convex and the fact that the gradient of a convex function's Moreau envelope is ρ^{-1} -Lipschitz. Therefore, let

$$L_{\mathcal{G}_{\rho\tilde{f}}} := L_{\nabla g} + \rho^{-1} \quad (84)$$

and we have

$$\left\| \mathcal{G}_{\rho\tilde{f}}(x) - \mathcal{G}_{\rho\tilde{f}}(y) \right\| \leq L_{\mathcal{G}_{\rho\tilde{f}}} \|x - y\|. \quad (85)$$

□

Furthermore, (75) suggests that

$$\mathcal{G}_{\rho\tilde{f}} = \nabla g + (\nabla M_\rho h) \circ \tilde{g}_\rho = \nabla g + (\nabla M_\rho h) \circ (Id - \rho \nabla g). \quad (86)$$

Hence,

$$\begin{aligned} Id - \rho \mathcal{G}_{\rho\tilde{f}} &= Id - \rho \nabla g - \rho (\nabla M_\rho h) \circ (Id - \rho \nabla g) \\ &= \tilde{g}_\rho - \rho (\nabla M_\rho h) \circ \tilde{g}_\rho \\ &= (Id - \rho (\nabla M_\rho h)) \circ \tilde{g}_\rho \end{aligned} \quad (87)$$

$$= \tilde{g}_\rho^{-1} \circ (\tilde{g}_\rho \circ (Id - \rho (\nabla M_\rho h))) \circ \tilde{g}_\rho \quad (88)$$

shows that $\tilde{g}_\rho^{-1} \circ (\tilde{g}_\rho \circ (Id - \rho (\nabla M_\rho h))) \circ \tilde{g}_\rho : \mathbb{R}^{p+1} \mapsto \mathbb{R}^{p+1}$ equals to $Id - \rho \mathcal{G}_{\rho\tilde{f}} : \mathbb{R}^{p+1} \mapsto \mathbb{R}^{p+1}$. Since $\tilde{g}_\rho$ is bijective, and that $\tilde{g}_\rho$ and $\tilde{g}_\rho^{-1}$ are continuous due to the globally Lipschitz property from Lemma 9, $\tilde{g}_\rho$ is a homeomorphism. Hence, $Id - \rho \mathcal{G}_{\rho\tilde{f}}$ and $\tilde{g}_\rho \circ (Id - \rho (\nabla M_\rho h))$ are topologically equivalent mappings via the homeomorphism $\tilde{g}_\rho$. Lemma 11 implies that $\mathcal{G}_{\rho\tilde{f}}$ is globally Lipschitz, which sufficiently implies by the Cauchy-Lipschitz theorem that the differential equation

$$\dot{x} := \frac{dx}{dt} = \mathcal{G}_{\rho\tilde{f}}(x) \quad (89)$$

has a unique solution for any given initial value condition. Thus, $\mathcal{G}_{\rho\tilde{f}}$ generates a unique flow under a given initial value condition.

The operator equations presented above can be understood as demonstrating how $\mathcal{G}_{\rho\tilde{f}}$ functions analogously to a gradient operator. Specifically, $Id - \rho \mathcal{G}_{\rho\tilde{f}}$ represents executing a descent operation in the $-\mathcal{G}_{\rho\tilde{f}}$ direction with a step size ρ . Similarly, $Id - \rho (\nabla M_\rho h)$ represents a single gradient descent step with ρ as the step size with objective function $M_\rho h$, the Moreau envelope of h ; while $\tilde{g}_\rho = Id - \rho \nabla g$ reflects a gradient descent step with objective function g , again with ρ as the step size. Equation (87) elucidates that a descent in the $-\mathcal{G}_{\rho\tilde{f}}$ direction is identical to first performing a one-step gradient descent on g , followed by $M_\rho h$; or performing gradient descents in a converse order yields a topologically equivalence via the homeomorphism $\tilde{g}_\rho$, as shown in (88).

In short summary, the approach based on linearization of the smooth term and the Moreau envelope enables us to build equivalence between identifying Clarke stationary points of the original nonsmooth objective function (101) and finding equilibria of the (unique) flow generated by $\mathcal{G}_{\rho\tilde{f}}$, as demonstrated in Lemma 10. The task of finding equilibria within a globally Lipschitz continuous flow, such as the $\mathcal{G}_{\rho\tilde{f}}$ flow, is well explored within mathematics, particularly in the realms of dynamical systems and numerical analysis (see, for example, [44, 2, 33, 27, 24]). Cauchy-Lipschitz theorem establishes the uniqueness

of solutions to initial value problems for globally Lipschitz continuous flows; while the existence of equilibria is a direct result of Brouwer fixed-point theorem. Numerical methods for dynamical systems, including methods for finding equilibria of the flow, are largely based on this uniqueness result. This is one reason that the vast majority of numerical methods in the context of dynamical systems require the flow to be globally Lipschitz. It is important to note that these numerical strategies, widely applied across dynamical systems, do not hinge on the flow being derived from a conservative field. As such, the process of formulating a potential function for $\mathcal{G}_{\rho\tilde{f}}$ is not a prerequisite for employing numerical techniques to determine its equilibria. This perspective underscores the versatility of numerical methods in dynamical systems in finding the equilibria of flows, regardless of the explicit existence of a potential function, a stance corroborated by various sources in the literature [44, 2, 33, 27, 24, 47, 45]. In this view, the construction of a potential function for $\mathcal{G}_{\rho\tilde{f}}$ is generally not necessary when deploying numerical analysis methods to find its equilibria.

In the context of nonlinear conjugate gradient algorithms for optimization, achieving global convergence on nonconvex objective functions that are globally Lipschitz-smooth implies that such methods can reliably find equilibria within the corresponding flow dynamics [47, 45]. These algorithms typically incorporate a line search step, which may use a surrogate objective function instead of the original. This surrogate can be a constructed potential, Lyapunov, or energy function, offering flexibility when finding the potential function for $\mathcal{G}_{\rho\tilde{f}}$ poses challenges [47, 9, 54].

When it is feasible to construct a potential function whose gradient with respect to x is $\mathcal{G}_{\rho\tilde{f}}$, the associated objective function and its gradient become more manageable, allowing for direct global convergence arguments. If constructing a potential function with respect to x for the $(\nabla M_\rho h) \circ \tilde{g}_\rho(x)$ term in (75) or $\nabla g \circ (Id - \rho(\nabla M_\rho h))$ in (88) is tractable, the objective function with gradient being (75) or $\tilde{g}_\rho \circ (Id - \rho(\nabla M_\rho h))$ can hence be easily constructed. Thus, arguments for global convergence for methods based on the objective function and its gradient directly follow to prove the global convergence of the numerical optimization algorithm when applied to the constructed potential function for $\mathcal{G}_{\rho\tilde{f}}$. We remark that $Id - \rho\mathcal{G}_{\rho\tilde{f}}$ and $\tilde{g}_\rho \circ (Id - \rho(\nabla M_\rho h))$ generate two topologically equivalent flows via homeomorphism $\tilde{g}_\rho$; thus, their equilibria can be transformed by $\tilde{g}_\rho$ and share the same stability. In the context of numerical optimization, this implies that a fixed point $\bar{x}$ for the mapping $Id - \rho\mathcal{G}_{\rho\tilde{f}}$ corresponds bijectively to a fixed point $\tilde{g}_\rho(\bar{x})$ for $\tilde{g}_\rho \circ (Id - \rho(\nabla M_\rho h))$. Characterized by the first-order optimality condition in optimization of smooth functions, or equivalently, the stationary condition in dynamical system,

$$\mathcal{G}_{\rho\tilde{f}}(\bar{x}) = \nabla g(\bar{x}) + (\nabla M_\rho h) \circ \tilde{g}_\rho(\bar{x}) = 0 \Leftrightarrow \nabla M_\rho h(\tilde{g}_\rho(\bar{x})) + \nabla g(\tilde{g}_\rho(\bar{x}) - \rho \nabla M_\rho h(\tilde{g}_\rho(\bar{x}))) = 0. \quad (90)$$

This approach is practical because the literature on first-order numerical optimization techniques frequently includes proofs of global convergence for methods that depend on the objective function and its gradient (for example, see [17, 43, 25, 13, 23]). Alternatively, construction of a potential function for $\mathcal{G}_{\rho\tilde{f}}$ is often not necessary due to the fact that

fixed-point methods finding equilibria for a flow mostly establish convergence properties based on Banach fixed point theorem. This theorem guarantees convergence through intrinsic flow properties, obviating the need for a potential function [Burden2001, 2, 1]. Conventionally, the use of line search based on the objective function and its gradient has been applied in some numerical methods to ensure global convergence. However, with the rapid growth of research in high-dimensional statistical machine learning and large-scale optimization, evaluations of the objective function often proven to be inefficient. Consequently, recent years have seen the exploration of two main alternatives. For instance, two different types of approaches for global convergent nonlinear conjugate gradient methods have been proposed without the conventional objective function-based line search procedure. One type of approach ensures global convergence by utilizing a line search mechanism that depends only on the nonlinear equation that generates the flow [16, 52, 53, 28]; that is, the gradient function for smooth optimization, or $\mathcal{G}_{\rho\tilde{f}}$ in our case. As an example, under the smoothness assumption, the first-order optimality condition for an exact line search often solves for α with the current value $x^{(k)}$ and the search direction $d^{(k)}$ from $\langle \mathcal{G}_{\rho\tilde{f}}(x^{(k)} + \alpha \cdot d^{(k)}), d^{(k)} \rangle = 0$, an equation dependent only on $\mathcal{G}_{\rho\tilde{f}}$ but not any surrogate objective function. From a practical perspective, this one-dimensional root finding problem can be carried out efficiently using the Brent root finding algorithm [7]. The other approach suggests achieving global convergence either without the need for line search [51, 8, 55, 62, 61, 65] or by meeting a condition related to the Zoutendijk condition to replace the Wolfe-Powell conditions of sufficient descent (Armijo) and curvature [39]. Additionally, in scenarios where the fulfillment of a sufficient descent (Armijo) condition is imperative, the formulation of a surrogate objective function becomes essential. Considering (75), where a surrogate objective is required for the line search phase, it could be formulated as:

$$\begin{aligned} & g(x) + (M_\rho h) \circ \tilde{g}_\rho(x) \\ &= g(x) + (M_\rho h) \circ \tilde{g}_\rho(x) + \text{constant} \\ &= g(x) + \langle \nabla g(x), \text{prox}_{\rho h}(x - \rho \nabla g(x)) - x \rangle + \frac{1}{2\rho} \|\text{prox}_{\rho h}(x - \rho \nabla g(x)) - x\|^2 \quad (91) \\ &\quad + h(\text{prox}_{\rho h}(x - \rho \nabla g(x))) + \text{constant} \end{aligned}$$

This formulation, denoted as (91), represents a quadratic approximation of g plus the nonsmooth term h , evaluated at $\text{prox}_{\rho h}(x - \rho \nabla g(x))$. This type of formulation has often been used for the line search step in previous studies [4, 29]. The addition of the term $-\langle \nabla g(x), x \rangle$ acts as a constant in (64), analogous to fixing the value of u as $\nabla g(x)$ for linearization. This constant term, $-\langle \nabla g(x), x \rangle$, doesn't alter the gradient of the Moreau envelope (66) or the proximal point (65), serving to frame the quadratic approximation of $g(\text{prox}_{\rho h}(x - \rho \nabla g(x)))$.

Evaluation of $\text{prox}_{\rho h}$ in (91) is tractable and efficient for many functions, such as the ℓ_1 norm commonly encountered in sparse statistical learning can be efficiently computed via the soft-thresholding function. Given that line search rules such as the Wolfe-Powell or

Armijo-Goldstein conditions require only the difference in the value of the objective function at two points to decide on the step size, the constant term in (91) can be disregarded. Subsequent global convergence arguments stem from the fixed-point theory analysis of the numerical methods deployed to find the equilibria of the $\mathcal{G}_{\rho\tilde{f}}$ flow. Another possible surrogate objective function inspired by the quadratic Lyapunov function for the $\mathcal{G}_{\rho\tilde{f}}$ flow could be $\frac{1}{2} \left\| \mathcal{G}_{\rho\tilde{f}} \right\|^2$, attains its minimal value 0 exactly at the $\mathcal{G}_{\rho\tilde{f}}$ flow's equilibria. This quadratic approach simplifies evaluation, but it may not offer insights into the potential function's landscape, potentially limiting the numerical algorithm's acceleration capabilities if such an algorithm uses the landscape information to ensure the sufficient descent (Armijo) condition. Therefore, formulating the surrogate objective function preserving the landscape of the original objective function as outlined in (91) is preferable.

Building on the above discussion, we introduce our practical proximal conjugate gradient framework in Algorithm 2.

Algorithm 2 Computationally Tractable Proximal Conjugate Gradient Update Scheme

- 1: Input: A fixed value of $\rho \in (0, \rho^{-1})$
 - 2: Calculate the proximal value $p^{(k)} := \text{prox}_{\rho^{-1}h}(x^{(k)} - \rho^{-1} \cdot \nabla g(x^{(k)}))$
 - 3: Calculate $\mathcal{G}_{\rho\tilde{f}}(x^{(k)})$: $s^{(k)} := \rho(x^{(k)} - p^{(k)})$
 - 4: $d^{(k)} := -s^{(k)} + \beta^{(k)} \cdot d^{(k-1)}$
 - 5: Line search to find $\alpha^{(k)}$ for the update $x^{(k+1)} := x^{(k)} + \alpha^{(k)}d^{(k)}$, if needed.
 - 6: Update $x^{(k+1)} := x^{(k)} + \alpha^{(k)}d^{(k)}$
-

In Algorithm 2, $\beta^{(k)}$ functions as the conjugate parameter. Unlike Algorithm 1, Algorithm 2 facilitates the update process without the need to compute $\nabla M_{\rho}f(x^{(k)})$. This adaptation is significantly valuable in practical scenarios, especially in statistical sparse learning challenges characterized by a complicated smooth component g alongside a simple nonsmooth convex component h . In such cases, computing $\text{prox}_{\rho h}$ is markedly more tractable and efficient than $\text{prox}_{\rho f}$. This approach is particularly beneficial for sparse statistical learning issues, where sparsity is commonly induced by an ℓ_1 penalty term.

4.3 Proximal Hager-Zhang [23] Conjugate Gradient

The nonlinear conjugate gradient method represents the pinnacle of first-order techniques for addressing smooth optimization challenges. Various versions of nonlinear conjugate gradient methods have been introduced, including the Fletcher-Reeves (FR) method [17], the modified Polak-Ribiere-Polyak (PRP+) method [43, 21], the Hestenes-Stiefel (HS) method [25], the Dai-Yuan (DY) method [13], and the Hager-Zhang (HZ) method [23]. These versions have all demonstrated global convergence with nonconvex globally Lipschitz-smooth objective functions. Among these, the Hager-Zhang conjugate gradient method is notable for delivering the best numerical performance on large-scale datasets, as indicated in pre-

vious research [22]. Building on this, having introduced our practical proximal conjugate gradient update mechanism in Algorithm 2, we aim to extend this approach by adapting the smooth Hager-Zhang nonlinear conjugate gradient method to its proximal version in Algorithm 3.

Algorithm 3 Proximal Hager-Zhang [23] Conjugate Gradient

- 1: **Input:** Initial point $x^{(0)}$; $g \in \mathcal{C}^{1,1}(\mathbb{R}^{p+1}, \mathbb{R})$; locally-Lipschitz, convex $h \in \mathcal{C}^{0,0}(\mathbb{R}^{p+1}, \mathbb{R})$; the smoothing parameter for the Moreau envelope $\rho \in (0, \rho^{-1})$; $k := 0$
 - 2: **Output:** p
 - 3: $k+ = 1$
 - 4: Calculate the gradient for g : $g^{(0)} := \nabla g(x^{(0)})$
 - 5: Calculate the proximal value $p^{(0)} := \text{prox}_{\rho, h}(x^{(0)} - \rho \cdot g^{(0)})$
 - 6: Calculate the gradient analog: $s^{(0)} := x^{(0)} - p^{(0)}$
 - 7: $d^{(0)} := -s^{(0)}$
 - 8: Perform the line search with $d^{(0)}$ with step size $\alpha^{(0)}$
 - 9: Update $x_1 := x^{(0)} + \alpha^{(0)}d^{(0)}$
 - 10: **while** not converged **do**
 - 11: $k+ = 1$
 - 12: Calculate the gradient for g : $g^{(k)} := \nabla g(x^{(k)})$
 - 13: Calculate the proximal value $p^{(k)} := \text{prox}_{\rho, h}(x^{(k)} - \rho \cdot g^{(k)})$
 - 14: Calculate the gradient analog: $s^{(k)} := x^{(k)} - p^{(k)}$
 - 15: $d^{(k)} := -s^{(k)} + \bar{\beta}^{(k)} \cdot d^{(k-1)}$
 - 16: Perform the line search with $d^{(k)}$ with step size $\alpha^{(k)}$ based on Wolfe-Powell conditions
 - 17: Update $x^{(k+1)} := x^{(k)} + \alpha^{(k)}d^{(k)}$
 - 18: Check for convergence
 - 19: **return** $p^{(k)}$
-

In Algorithm 3, Hager-Zhang's conjugate parameter $\bar{\beta}^{(k)}$ is defined as [23]:

$$\begin{aligned}
 y^{(k)} &:= s^{(k+1)} - s^{(k)} \\
 \beta^{(k)} &:= \frac{1}{\langle d^{(k)}, y^{(k)} \rangle} \cdot \left\langle y^{(k)} - 2 \frac{\|y^{(k)}\|^2}{\langle d^{(k)}, y^{(k)} \rangle} d^{(k)}, s^{(k+1)} \right\rangle \\
 \eta^{(k)} &:= -\frac{1}{\|d^{(k)}\| \min \{\eta, \|s^{(k)}\|\}} \\
 \bar{\beta}^{(k)} &:= \max \left\{ \beta^{(k)}, \eta^{(k)} \right\}
 \end{aligned}$$

It was proven that if the line search step in Algorithm 3 satisfies Wolfe-Powell conditions and the gradient is globally Lipschitz, Hager-Zhang conjugate gradient achieves

global convergence finding a stationary point for a smooth nonconvex objective function. In a dynamical system view, this corresponds to the global attraction property of the trajectory of the numerical algorithm to find equilibria for globally Lipschitz flows. Lemma 11 implies that $\mathcal{G}_{\rho\tilde{f}}$, or $s^{(k)}$ in Algorithm 3, are globally Lipschitz. Thus, by Lemma 10, Algorithm 3 yields the Clarke stationary point of f . Based on the arguments in Section 4.2, if the potential function for $\mathcal{G}_{\rho\tilde{f}}$ is tractable to construct, the Wolfe-Powell line search in Algorithm 3 can be carried out using the potential function of $\mathcal{G}_{\rho\tilde{f}}$ as the surrogate objective function; alternatively, an exact line search can be carried out by finding α that satisfies $\langle \mathcal{G}_{\rho\tilde{f}}(x^{(k)} + \alpha \cdot d^{(k)}), d^{(k)} \rangle = 0$ — such an exact line search can usually be carried out efficiently using Brent’s method to find a root of a one-dimensional equation in $\mathbb{R}_{>0}$ [7]. Furthermore, the descent property of $d^{(k)}$ was shown by [23] independent of the line searches, which guarantees that $\langle \mathcal{G}_{\rho\tilde{f}}(x^{(k)} + \alpha \cdot d^{(k)}), d^{(k)} \rangle = 0$ has a positive root. Moreover, another line search to ensure global convergence can be carried out by backtracking to find $\alpha^{(k)}$ satisfying

$$-\langle \mathcal{G}_{\rho\tilde{f}}(x^{(k)} + c_1 \cdot \alpha^{(k)} d^{(k)}), d^{(k)} \rangle \geq c_1 c_2 \cdot \alpha^{(k)} \|d^{(k)}\|^2, \quad (92)$$

where $c_1, c_2 \in \mathbb{R}_{>0}$ are constant to be chosen. When $\mathcal{G}_{\rho\tilde{f}}$ is pseudo-monotone in the sense of Karamardian [30], since the global Lipschitz property was established for $\mathcal{G}_{\rho\tilde{f}}$ in Lemma 11, global convergence was proven for this backtracking line search method [16]. We conclude this section with the observation that certain conjugate gradient methods obviate the need for line search procedures by determining the step size directly from $s^{(k)}$ and $d^{(k)}$, as exemplified in [8].

5 Optimizing Algorithm and Prediction for Penalized q Gaussian Likelihood Problems

5.1 Problem Formulation

Using the q Gaussian distribution to model the data will undoubtedly enhance the robustness towards the underlying distributional assumption and outliers. However, unlike the Gaussian distribution, two independent q Gaussian random vectors are not jointly q Gaussian. Thus, we take the following approach to model the data. Let $\mathbf{X}_{\text{train}} \in \mathbb{R}^{n_{\text{train}} \times (p+1)}$, $y_{\text{train}} \in \mathbb{R}^{n_{\text{train}}}$ denote the training design matrix and outcome, $\mathbf{X}_{\text{val}} \in \mathbb{R}^{n_{\text{val}} \times (p+1)}$, $y_{\text{val}} \in \mathbb{R}^{n_{\text{val}}}$ denote the validation design matrix and outcome, and $\mathbf{X}_{\text{test}} \in \mathbb{R}^{n_{\text{test}} \times (p+1)}$, $y_{\text{test}} \in \mathbb{R}^{n_{\text{test}}}$ denote the testing design matrix and outcome. Let

$$\mathbf{X} := [\mathbf{X}_{\text{train}}^T, \mathbf{X}_{\text{val}}^T, \mathbf{X}_{\text{test}}^T]^T \in \mathbb{R}^{n \times (p+1)} \quad (93)$$

denote the design matrix for the entire dataset, and let

$$y := [y_{\text{train}}^T, y_{\text{val}}^T, y_{\text{test}}^T]^T \in \mathbb{R}^n \quad (94)$$

denote the outcome for the entire dataset. Instead of assuming the q Gaussian distribution for the training, validation and testing set separately, we assume that

$$y \sim q\text{Gaussian}(q, \mathbf{X}\theta, \Sigma), \quad (95)$$

where $\theta \in \mathbb{R}^{p+1}$ denotes the coefficients for regression, and Σ denotes the characteristic/scale matrix for the entire data. Clearly,

$$\begin{aligned} \mathbf{X}_{\text{train}} &= [I_{n_{\text{train}} \times n_{\text{train}}}, 0_{n_{\text{train}} \times n_{\text{val}}}, 0_{n_{\text{train}} \times n_{\text{test}}}] \mathbf{X} \\ y_{\text{train}} &= [I_{n_{\text{train}} \times n_{\text{train}}}, 0_{n_{\text{train}} \times n_{\text{val}}}, 0_{n_{\text{train}} \times n_{\text{test}}}] y \end{aligned}$$

implies that

$$y_{\text{train}} \sim q\text{Gaussian}(q_{\text{train}}, \mathbf{X}_{\text{train}}\theta, \Sigma_{\text{train}}) \quad (96)$$

where by the linear mapping closeness property 2,

$$\Sigma_{\text{train}} = [I_{n_{\text{train}} \times n_{\text{train}}}, 0_{n_{\text{train}} \times n_{\text{val}}}, 0_{n_{\text{train}} \times n_{\text{test}}}] \Sigma [I_{n_{\text{train}} \times n_{\text{train}}}, 0_{n_{\text{train}} \times n_{\text{val}}}, 0_{n_{\text{train}} \times n_{\text{test}}}]^T \quad (97)$$

is the $n_{\text{train}} \times n_{\text{train}}$ block diagonal matrix of Σ corresponding to the training data. By (2),

$$\frac{2}{1 - q_{\text{train}}^{-1}} - n_{\text{train}} = \frac{2}{1 - q^{-1}} - n, \quad (98)$$

which implies that

$$\frac{1}{q_{\text{train}} - 1} - n_{\text{train}} = \frac{1}{q - 1} - n. \quad (99)$$

(99) allows us to recover q from the training procedure. Above formulas for the training data and parameters can trivially be applied to the validation and the testing data and parameters; thus, validation and test can be carried out easily from the model built from the training data.

For q -correlated data, often times, the q -correlation structure is inferred or given prior to the model fitting; thus, we assume that the q -correlation structure is given as Ψ and we estimate the volatility / dispersion / scale parameter $\sigma^2 > 0$ such that

$$\Sigma = \sigma^2 \Psi. \quad (100)$$

Trivially, Ψ_{train} is the block diagonal matrix of Ψ corresponding to the training data and $\Sigma_{\text{train}} = \sigma^2 \Psi_{\text{train}}$.

We are now ready to formulate our likelihood loss function. To utilize q Gaussian distribution to model the q -correlated observations, we estimate the value of q such that q is allowed to vary, and the model will thus be more robust towards a wide class of distributions. Therefore, we choose to build the model using (20), since the dispersion

matrix Λ (31) depends on q . We formulate our maximization of our log-likelihood function as the following from (96) and (20):

$$\begin{aligned}
 & \underset{q_{\text{train}} \in (1, 1 + \frac{2}{n_{\text{train}}}), \theta \in R^{p+1}, \sigma^2 \in R_{>0}}{\arg \max} \log \left(\frac{1}{|\sigma^2 \Psi_{\text{train}}|^{1/2}} \right) \\
 & \cdot \frac{\Gamma \left(\frac{1}{q_{\text{train}} - 1} \right)}{\Gamma \left(\frac{1}{q_{\text{train}} - 1} - \frac{n_{\text{train}}}{2} \right)} \cdot \left(\frac{2}{q_{\text{train}} - 1} - n_{\text{train}} \right)^{-\frac{n_{\text{train}}}{2}} \\
 & \cdot \left(1 + \left(\frac{2}{q_{\text{train}} - 1} - n_{\text{train}} \right)^{-1} \cdot \left\langle y_{\text{train}} - \mathbf{X}_{\text{train}} \theta, (\sigma^2 \Psi_{\text{train}})^{-1} (y_{\text{train}} - \mathbf{X}_{\text{train}} \theta) \right\rangle \right)^{\frac{1}{1 - q_{\text{train}}}}.
 \end{aligned}$$

To address the high-dimensional data concerns, Oracle penalties are incorporated to carry out variable selection. To penalize the log-likelihood loss function to achieve variable selection, we formulate the following problem:

$$\begin{aligned}
 & \underset{q_{\text{train}} \in (1, 1 + \frac{2}{n_{\text{train}}}), \theta \in R^{p+1}, \sigma^2 \in R_{>0}}{\arg \min} \\
 & - \log \left(\frac{1}{|\sigma^2 \Psi_{\text{train}}|^{1/2}} \cdot \frac{\Gamma \left(\frac{1}{q_{\text{train}} - 1} \right)}{\Gamma \left(\frac{1}{q_{\text{train}} - 1} - \frac{n_{\text{train}}}{2} \right)} \cdot \left(\frac{2}{q_{\text{train}} - 1} - n_{\text{train}} \right)^{-\frac{n_{\text{train}}}{2}} \right. \\
 & \cdot \left. \left(1 + \left(\frac{2}{q_{\text{train}} - 1} - n_{\text{train}} \right)^{-1} \cdot \sigma^{-2} \cdot \left(\left\langle y_{\text{train}} - \mathbf{X}_{\text{train}} \theta, \Psi_{\text{train}}^{-1} (y_{\text{train}} - \mathbf{X}_{\text{train}} \theta) \right\rangle + 2n_{\text{train}} \sum_{j=2}^{p+1} w(\theta_j) \right) \right)^{\frac{1}{1 - q_{\text{train}}}} \right) \\
 & \Leftrightarrow \underset{q_{\text{train}} \in (1, 1 + \frac{2}{n_{\text{train}}}), \theta \in R^{p+1}, \sigma^2 \in R_{>0}}{\arg \min} \frac{n}{2} \log \sigma^2 - \log \Gamma \left(\frac{1}{q_{\text{train}} - 1} \right) + \log \Gamma \left(\frac{1}{q_{\text{train}} - 1} - \frac{n_{\text{train}}}{2} \right) \\
 & + \frac{n_{\text{train}}}{2} \log \left(\frac{2}{q_{\text{train}} - 1} - n_{\text{train}} \right) + \frac{1}{q_{\text{train}} - 1} \log \left(1 + \left(\frac{2}{q_{\text{train}} - 1} - n_{\text{train}} \right)^{-1} \cdot \sigma^{-2} \right. \\
 & \cdot \left. \left(\left\langle y_{\text{train}} - \mathbf{X}_{\text{train}} \theta, \Psi_{\text{train}}^{-1} (y_{\text{train}} - \mathbf{X}_{\text{train}} \theta) \right\rangle + 2n_{\text{train}} \sum_{j=2}^{p+1} w(\theta_j) \right) \right) \tag{101}
 \end{aligned}$$

In the above formulated problem, w is the Oracle penalty function, and we are not to penalize the intercept term. The $2n_{\text{train}}$ multiplier is to ensure that the penalization effect is consistent with the number of training observations. We choose to put the penalty term together with the quadratic term without the variance scale parameter σ^2 for two reasons: first, the optimization problem is more tractable under such problem formulation; second, we do not wish to let the value of σ^2 perturb the degree of penalization. Comparing

to penalized the log-likelihood directly, we choose to penalize the quadratic component directly as it is more tractable. It was shown that doing so will preserve Oracle properties [40] of penalized estimators.

For the optimization procedure, we will proceed in a blockwise manner; i.e., we will optimize $q_{\text{train}}, \theta, \sigma^2$ separately in each iteration. More details will be given in the following subsections.

5.2 Minimizing with respect to q_{train} and σ^2

With all the other parameters fixed, the sub-problem to minimize with respect to σ^2 is

$$\begin{aligned} \arg \min_{\sigma^2 \in R_{>0}} \frac{n}{2} \log \sigma^2 + \frac{1}{q_{\text{train}} - 1} \log \left(1 + \left(\frac{2}{q_{\text{train}} - 1} - n_{\text{train}} \right)^{-1} \cdot \sigma^{-2} \right. \\ \left. \cdot \left(\langle y_{\text{train}} - \mathbf{X}_{\text{train}} \theta, \Psi_{\text{train}}^{-1} (y_{\text{train}} - \mathbf{X}_{\text{train}} \theta) \rangle + 2n_{\text{train}} \sum_{j=2}^{p+1} w(\theta_j) \right) \right) \end{aligned} \quad (102)$$

which has a smooth objective function with respect to σ^2 . The first-order optimality condition

$$\begin{aligned} \frac{n}{2} = \frac{1}{q_{\text{train}} - 1} \\ \cdot \frac{\left(\frac{2}{q_{\text{train}} - 1} - n_{\text{train}} \right)^{-1} \cdot \left(\langle y_{\text{train}} - \mathbf{X}_{\text{train}} \theta, \Psi_{\text{train}}^{-1} (y_{\text{train}} - \mathbf{X}_{\text{train}} \theta) \rangle + 2n_{\text{train}} \sum_{j=2}^{p+1} w(\theta_j) \right)}{\sigma^2 + \left(\frac{2}{q_{\text{train}} - 1} - n_{\text{train}} \right)^{-1} \cdot \left(\langle y_{\text{train}} - \mathbf{X}_{\text{train}} \theta, \Psi_{\text{train}}^{-1} (y_{\text{train}} - \mathbf{X}_{\text{train}} \theta) \rangle + 2n_{\text{train}} \sum_{j=2}^{p+1} w(\theta_j) \right)} \end{aligned}$$

implies that the optimal value for the subproblem (102) takes minimizer

$$\begin{aligned} \bar{\sigma}^2 = \left(\frac{1}{q_{\text{train}} - 1} / \frac{n}{2} - 1 \right) \cdot \left(\frac{2}{q_{\text{train}} - 1} - n_{\text{train}} \right)^{-1} \\ \cdot \left(\langle y_{\text{train}} - \mathbf{X}_{\text{train}} \theta, \Psi_{\text{train}}^{-1} (y_{\text{train}} - \mathbf{X}_{\text{train}} \theta) \rangle + 2n_{\text{train}} \sum_{j=2}^{p+1} w(\theta_j) \right) > 0, \end{aligned} \quad (103)$$

which is feasible. The feasible set for q_{train} is $\left(1, 1 + \frac{2}{n_{\text{train}}} \right)$, in this view, when n_{train} is large, the numerical stability will be an issue if minimization is carried out with respect to q_{train} directly. Thus, we choose to minimize with respect to $\frac{1}{q_{\text{train}} - 1} \in \left(\frac{n_{\text{train}}}{2}, \infty \right)$.

First of all, we are to prove that such minimization is feasible.

Lemma 12. *The objective function (101) has a local minimizer in $\left(\frac{n_{\text{train}}}{2}, \infty \right)$ with respect to $\frac{1}{q_{\text{train}} - 1}$.*

Proof. Since the objective function (101) is continuous and smooth with respect to $\frac{1}{q_{\text{train}}-1}$, we only need to analyze the derivative when $\frac{1}{q_{\text{train}}-1} \searrow 0$ and $\frac{1}{q_{\text{train}}-1} \rightarrow \infty$.

$\frac{1}{q_{\text{train}}-1} \rightarrow \infty$:

Stirling's formula states that

$$\lim_{x \rightarrow \infty} \frac{\Gamma(x)}{\sqrt{\frac{2\pi}{x}} \left(\frac{x}{e}\right)^x (1 + O(x^{-1}))} = 1. \quad (104)$$

Thus,

$$\lim_{\frac{1}{q_{\text{train}}-1} \rightarrow \infty} \frac{\Gamma\left(\frac{1}{q_{\text{train}}-1}\right)}{\Gamma\left(\frac{1}{q_{\text{train}}-1} - \frac{n_{\text{train}}}{2}\right)} \cdot \left(\frac{2}{q_{\text{train}}-1} - n_{\text{train}}\right)^{-\frac{n_{\text{train}}}{2}} = 1 \quad (105)$$

then

$$\lim_{\frac{1}{q_{\text{train}}-1} \rightarrow \infty} -\log \left(\frac{\Gamma\left(\frac{1}{q_{\text{train}}-1}\right)}{\Gamma\left(\frac{1}{q_{\text{train}}-1} - \frac{n_{\text{train}}}{2}\right)} \cdot \left(\frac{2}{q_{\text{train}}-1} - n_{\text{train}}\right)^{-\frac{n_{\text{train}}}{2}} \right) = 0. \quad (106)$$

We also have

$$\begin{aligned} & \frac{1}{q_{\text{train}}-1} \log \left(1 + \left(\frac{2}{q_{\text{train}}-1} - n_{\text{train}} \right)^{-1} \cdot \sigma^{-2} \right. \\ & \quad \left. \cdot \left(\left\langle y_{\text{train}} - \mathbf{X}_{\text{train}} \theta, \Psi_{\text{train}}^{-1} (y_{\text{train}} - \mathbf{X}_{\text{train}} \theta) \right\rangle + 2n_{\text{train}} \sum_{j=2}^{p+1} w(\theta_j) \right) \right) \\ & = O \left(\left(\frac{1}{q_{\text{train}}-1} \right) / \log \left(\frac{1}{q_{\text{train}}-1} \right) \right), \end{aligned}$$

which implies that this term will goes to infinity as $\frac{1}{q_{\text{train}}-1} \rightarrow \infty$. Thus, the objective function (101) goes to infinity as $\frac{1}{q_{\text{train}}-1} \rightarrow \infty$.

$\frac{1}{q_{\text{train}}-1} \searrow \frac{n_{\text{train}}}{2}$:

Since $\Gamma\left(\frac{1}{q_{\text{train}}-1} - \frac{n_{\text{train}}}{2}\right) \rightarrow \infty$ as $\frac{1}{q_{\text{train}}-1} \searrow \frac{n_{\text{train}}}{2}$.

The penalized log-likelihood involving $\frac{1}{q_{\text{train}}-1}$ can be simplified as

$$\begin{aligned} & -\log \frac{\Gamma\left(\frac{1}{q_{\text{train}}-1}\right)}{\Gamma\left(\frac{1}{q_{\text{train}}-1} - \frac{n_{\text{train}}}{2}\right)} \cdot \left(\left(\frac{2}{q_{\text{train}}-1} - n_{\text{train}} \right) \right. \\ & \quad \left. + \left\langle y_{\text{train}} - \mathbf{X}_{\text{train}} \theta, (\sigma^2 \Psi_{\text{train}})^{-1} (y_{\text{train}} - \mathbf{X}_{\text{train}} \theta) \right\rangle + 2n_{\text{train}} \sum_{j=2}^{p+1} w(\theta_j) \right)^{\frac{1}{1-q_{\text{train}}}} \rightarrow \infty \end{aligned}$$

as $\frac{1}{q_{\text{train}}-1} \searrow \frac{n_{\text{train}}}{2}$. Thus, the subproblem to minimize with respect to $\frac{1}{q_{\text{train}}-1}$ is coercive on $(\frac{n_{\text{train}}}{2}, \infty)$. Coercivity implies that any minimizing sequence $\left\{ \left(\frac{1}{q_{\text{train}}-1} \right)_j \right\}$ must be contained within a bounded subset of $(\frac{n_{\text{train}}}{2}, \infty)$. Thus, Bolzano–Weierstrass theorem implies the existence of a convergent subsequence. Let $\left\{ \left(\frac{1}{q_{\text{train}}-1} \right)_{j_k} \right\}$ be one such subsequence, and let $\overline{\left(\frac{1}{q_{\text{train}}-1} \right)}$ be its limit. Since the subproblem has a continuous objective function with respect to $\frac{1}{q_{\text{train}}-1}$, the objective function is lower-semicontinuous and the value of the objective function at $\overline{\left(\frac{1}{q_{\text{train}}-1} \right)}$ is less than or equal to the value of the objective function at $\left(\frac{1}{q_{\text{train}}-1} \right)_{j_k}$ for all $k = 1, 2, \dots, \infty$. Thus, since $\left\{ \left(\frac{1}{q_{\text{train}}-1} \right)_j \right\}$ is a minimizing sequence, the value of the objective function at $\overline{\left(\frac{1}{q_{\text{train}}-1} \right)}$ is less than or equal to the infimum of the objective function on $(\frac{n_{\text{train}}}{2}, \infty)$. Hence, since the entire minimizing sequence is contained in $(\frac{n_{\text{train}}}{2}, \infty)$, $\overline{\left(\frac{1}{q_{\text{train}}-1} \right)} \in (\frac{n_{\text{train}}}{2}, \infty)$ solves the subproblem of minimizing with respect to $\frac{1}{q_{\text{train}}-1}$.

Unlike σ^2 , the minimizer for $\frac{1}{q_{\text{train}}-1}$ is not in closed form, and the evaluation of the derivative with respect to $\frac{1}{q_{\text{train}}-1}$ can not be carried out efficiently. Thus, we apply Brent’s line-search method to optimize the $\frac{1}{q_{\text{train}}-1}$ subproblem [7]. $\square$

5.3 Minimizing with respect to θ

Minimizing with respect to θ involves a nonconvex smooth function and a convex nonsmooth function, which is termed a composite problem. In Section 4.3, we developed a proximal conjugate gradient algorithm for such composite optimization.

In this part, we will establish an important remark regarding the Oracle penalty. The subproblem we are to minimize with respect to θ is

$$\begin{aligned}
 & \arg \min_{\theta \in R^{p+1}} \langle y_{\text{train}} - \mathbf{X}_{\text{train}}\theta, \Psi_{\text{train}}^{-1}(y_{\text{train}} - \mathbf{X}_{\text{train}}\theta) \rangle + 2n_{\text{train}} \sum_{j=2}^{p+1} w(\theta_j) \\
 & \Leftrightarrow \arg \min_{\theta \in R^{p+1}} \frac{1}{2n_{\text{train}}} \langle y_{\text{train}} - \mathbf{X}_{\text{train}}\theta, \Psi_{\text{train}}^{-1}(y_{\text{train}} - \mathbf{X}_{\text{train}}\theta) \rangle + \sum_{j=2}^{p+1} w(\theta_j) \\
 & \Leftrightarrow \arg \min_{\theta \in R^{p+1}} \frac{1}{2n_{\text{train}}} \langle \theta, \mathbf{X}_{\text{train}}^T \Psi_{\text{train}}^{-1} \mathbf{X}_{\text{train}} \theta \rangle - 2 \langle y_{\text{train}}, \Psi_{\text{train}}^{-1} \mathbf{X}_{\text{train}} \theta \rangle + \sum_{j=2}^{p+1} w(\theta_j) \quad (107)
 \end{aligned}$$

w can be chosen as oracle penalties such as SCAD/MCP penalties. And it has been shown that both SCAD / MCP penalties admit a difference-of-convex decomposition to a first-

order smooth concave term plus λ times ℓ_1 penalty. The quadratic loss function is clearly convex and smooth. This justifies our assumption for the objective function. To carry out the proximal Hager-Zhang conjugate gradient method proposed in Section 4.3, we need to calculate $L_{\nabla g}$, the L -smoothness constant for the smooth component. Previous work suggests $L_{\nabla g} = \max \left\{ \max \text{eigenvalue of } \frac{1}{n_{\text{train}}} \mathbf{X}_{\text{train}}^T \Psi_{\text{train}}^{-1} \mathbf{X}_{\text{train}}, c_{\text{penalty}} \right\}$, where c_{penalty} is the L -smoothness constant for the smooth component of the penalty, which will be $\frac{1}{a-1}$ for SCAD and $\frac{1}{\gamma}$ for MCP [63].

Remark 13. For high dimensional data, often times, the number of covariates exceeds the number of observations; i.e, $\text{null}(\mathbf{X}_{\text{train}}) \neq \emptyset$. Both SCAD/MCP penalties take constant values in $B_{\infty}(0, c)^1$; where $c = a\lambda$ for SCAD and $c = \gamma\lambda$ for MCP. Given any stationary point $\bar{\theta}$ in the nonempty solution set defined by $\mathbf{X}_{\text{train}}^T \Psi_{\text{train}}^{-1} \mathbf{X}_{\text{train}} - \mathbf{X}_{\text{train}}^T y_{\text{train}} = 0$. For the set $\bar{\theta} + \text{null}(\mathbf{X}_{\text{train}}) \setminus B_{\infty}(0, c)$, each point in the relative interior (which is nonempty) of this set is a Clarke stationary point, which implies that any algorithm with a starting point in this set will converge in 0 steps. This might pose an issue for signal recovery, since $\text{null}(\mathbf{X}_{\text{train}})$ is a vector subspace and some points can be very far from the origin.

Remark 14. In view of the subproblem with respect to θ , it is trivial that the minimizer for (107) does not depend on the other parameters, which are q and σ^2 . Since the q Gaussian distribution is a generalization for all bell curve distributions, the estimation of the central trend using the maximum likelihood principle for bell curve distributions is equivalent to minimize a quadratic function, which has a breakdown point of 0.

Taking into account the optimization subproblem with respect to θ , it is evident that the solution to (107) remains unaffected by the other parameters, namely q and σ^2 . Given that the q Gaussian distribution extends the framework of bell curve distributions, the problem (107) implies that estimating the central trend through the maximum likelihood principle for all bell curve distributions is equivalent to minimizing a quadratic function of the central trend, therefore characterized by a breakdown point of 0.

5.4 Prediction for y_{test}

To show how prediction can be made, we will show the methods to predict y_{test} using the trained model in this subsection. The same method applies for validation when predictions on y_{val} are needed or to predict any new data based on the trained model. Since the data are mutually q Gaussian, (99) implies that

$$\frac{1}{q_{\text{train}} - 1} - n_{\text{train}} = \frac{1}{q_{\text{val}} - 1} - n_{\text{val}} = \frac{1}{q_{\text{test}} - 1} - n_{\text{test}} = \frac{1}{q - 1} - n, \quad (108)$$

which will be used to recover the value of the shape parameter $q_{\text{val}}, q_{\text{test}}$. Note that when n_{new} data points are introduced, the total number of observations n changes from $n_{\text{train}} +$

1. $B_{\infty}(0, c)$ denotes the open ball in uniform norm in the corresponding space, centered at the origin with radius c .

$n_{\text{val}} + n_{\text{test}}$ to $n_{\text{train}} + n_{\text{val}} + n_{\text{test}} + n_{\text{new}}$, thus, the value of q will change for the entire dataset. However, q_{train} stays the same; thus, we suggest inferring the shape parameter for each data set based on q_{train} directly using the equation above. With q calculated, it is straightforward to estimate the q -variance-covariance matrix $\mathbb{E}_q \left[(y - \mathbf{X}\theta) (y - \mathbf{X}\theta)^T \right]$ based on (33); or, *if existing*, the variance-covariance matrix $\mathbb{E} \left[(y - \mathbf{X}\theta) (y - \mathbf{X}\theta)^T \right]$ based on (34).

6 Conclusion and Discussion

This paper explores the field of statistical sparse learning, focusing on modeling correlated data through the lens of maximizing Tsallis entropy. It addresses the limitations inherent in the conventional Gaussian distribution, notably its lack of robustness towards outliers and underlying shape assumptions, by advocating for the q Gaussian distribution. This distribution, derived from Tsallis entropy maximization, represents a novel approach to handling correlated data and heterogeneity — elements frequently encountered in biostatistical contexts involving genetic and longitudinal studies.

This paper encompasses a re-derived probability density function for the multivariate q Gaussian distribution based on Tsallis entropy maximization. Statistical modeling based on the derived density paves the way for the analysis of correlated data and heterogeneity and enables variable selection. Furthermore, we have developed an innovative framework capable of converting any numerical method, originally designed to identify equilibria in flows, into a tool for tackling composite optimization problems that are prevalent in statistical sparse learning. By applying this framework to the Hager-Zhang conjugate gradient algorithm, we have crafted an effective and stable algorithm tailored to the challenges of sparse statistical learning. Given the abundance of methods for numerically identifying equilibria for globally Lipschitz flows, our approach significantly broadens the arsenal of techniques available to address sparse statistical learning optimization challenges.

In conclusion, our research positions the q Gaussian distribution, underpinned by maximizing Tsallis entropy, as a robust and adaptable alternative to Gaussian-based methodologies in statistical sparse learning on correlated data. This breakthrough not only confronts the traditional limitation of Gaussian assumptions, but also paves the way for expanded investigation into Tsallis entropy-maximizing distributions, particularly within the domain of biostatistics and allied disciplines.

Future directions for research include the exploration of the log-linear model through the lens of Tsallis entropy maximization, akin to approaches previously based on Shannon's entropy. Moreover, the study of the phenomenon called *volatility smirk* in financial return data may benefit from employing the log- q Gaussian distribution — a transformation of the q Gaussian distribution, which can provide deeper insights into the nuances of financial markets. Additionally, in the field of statistical computing research, our framework that transforms numerical methods for identifying flow equilibria into algorithms for solving

composite optimization problems opens numerous avenues for future research, especially in a sparse learning context.

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